Renewable and Sustainable Energy Reviews 143 (2021) 110933

Contents lists available at ScienceDirect

Renewable and Sustainable Energy Reviews

journal homepage: http://www.elsevier.com/locate/rser

# Solutions to obstacles in the commercialization of room-temperature magnetic refrigeration

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ARTICLE INFO

*Keywords:*
Magnetic refrigeration
Thermodynamic cycles
Fully solid-state MR cycle
Multi-caloric refrigeration cycle
Heat transfer enhancement
Application

ABSTRACT

Most existing room temperature magnetic refrigeration (MR) prototypes are based on the concept of an active magnetic regenerator (AMR). However, for these MR prototypes, three obstacles, i.e. theoretical limit of the MR thermodynamic cycle, low operating frequency, and large irreversible loss during heat regeneration, limit the enhancement of their temperature span and cooling capacity, and further restrict their commercial application. In this paper, the solutions to these obstacles are reviewed from the perspectives of MR thermodynamic cycles and heat transfer enhancement during heat regeneration. With respect to MR cycles, the future trend is likely to be fully solid-state MR cycle and multi-caloric refrigeration cycle. With regard to heat transfer enhancement, the three methods exhibit good practical prospects, namely using liquid metals or nanofluids as the heat transfer fluid, shaping a magnetocaloric material (MCM) into an enhanced heat transfer structure, and inserting materials with high thermal conductivity in the MCM. Moreover, room-temperature MR applications in the cold-storage device, heat pump and electric vehicle air conditioning are reviewed. Small cooling capacity devices are the primary application targets of room-temperature MR, such as the wine cooler, domestic dehumidifier, and portable personal air conditioning.

## 1. Introduction

In the modern society, refrigeration has become an increasingly essential technology, with applications ranging from the food industry and health care to the energy sector [1]. Currently, the predominant refrigeration technology is vapor compression (VC) refrigeration, which is characterized by low energy efficiency, and the use of environmentally harmful gas refrigerants [2]. The efficiency of VC refrigeration just reaches 50% of the Carnot efficiency [3], resulting in significant energy waste. Furthermore, refrigerant hydrofluorocarbons (HFCs) have a global warming potential (GWP) larger than 1000 [4], whose release powerfully contributes to global warming. Therefore, an alternative refrigeration technology that is energy efficient and environmentally friendly is urgently required.

Magnetic refrigeration (MR) is based on the magnetocaloric effect (MCE) of a magnetocaloric material (MCM). When an MCM is exposed to an increased magnetic field, its temperature increases. The opposite occurs when the magnetic field is decreased or removed, and the MCM temperature decreases. Due to the reversibility of the MCM temperature change, an efficiency of up to 70% of the Carnot efficiency can be obtained through MR [5]. Moreover, MR uses MCMs with zero-GWP as refrigerants, making it a truly green refrigeration technology [6]. Based on these positive aspects, MR is considered as a promising alternative to VC refrigeration [7].

In the early days, MR was primarily used to obtain extremely low temperature of milli-Kelvin [8]. The development of room-temperature MR began in 1976 when Brown built the first room-temperature MR prototype based on a regenerative cycle [9]. In this prototype, Gd was used as the MCM, and ethanol-water was used as the heat transfer fluid (HTF). Under a 7 T superconducting magnet, a 47 K temperature span (usually defined as the temperature difference between the heat source and heat sink) was achieved. Brown's study was innovative because it demonstrated not only that MR is feasible at room temperature, but also that a regenerative cycle can produce a temperature lift much larger than the adiabatic temperature change (MCM temperature change when magnetized/demagnetized adiabatically [10]) [11].

Contrary to the conventional passive regeneration adopted by Brown [9], in 1982, Barclay and Steyert presented the concept of an active magnetic regenerator (AMR) cycle [12], which subsequently formed the

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https://doi.org/10.1016/j.rser.2021.110933
Received 4 September 2020; Received in revised form 2 February 2021; Accepted 3 March 2021
Available online 11 March 2021
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**Abbreviations**

| | |
|---|---|
| AMR | Active magnetic regenerator |
| BRIRE | Baotou research institute of rare earths |
| CES | Consumer electronics show |
| CHEX | Cold end heat exchanger |
| COP | Coefficient of performance |
| Gd | Gadolinium |
| GE | General electric |
| GSHE | Ground source heat exchanger |
| GWP | Global warming potential |
| HFCs | Hydrofluorocarbons |
| HHEX | Hot end heat exchanger |
| HTCM | High thermal conductivity material |
| HTF | Heat transfer fluid |
| LTCM | Low thermal conductivity material |
| MCE | Magnetocaloric effect |
| MCM | Magnetocaloric material |
| MR | Magnetic refrigeration |
| MUR | Micro-unit regeneration |
| PHE | Plate heat exchanger |
| SCP | Specific cooling power |
| VC | Vapor compression |

theoretical basis for almost all MR prototypes. The operating process of an AMR cycle is shown in Fig. 1, and consists of four steps:

1) Magnetization: the MCM in the AMR is magnetized and its temperature increases.
2) Cold to hot flow: the HTF is pushed from the cold end through the AMR to the hot end where the heat received from the MCM is released to the surroundings.
3) Demagnetization: the MCM is demagnetized and its temperature decreases.
4) Hot to cold flow: the HFT is pushed from the hot end to cold end. The HTF is cooled down by MCM in the AMR, and then absorbs heat from the surroundings.

The operating processes of the AMR indicates that it has a distinctive feature differentiates it from the passive regenerator. That is, the MCM in AMR has the dual function of refrigerant and regenerative medium [13]. When acting as a refrigerant, the MCM exhibits the MCE. When the MCM serves as a regenerative medium, a HFT is pumped back and forth through the MCM, thereby establishing a temperature gradient along the length of the AMR.

After the introduction of AMR, room-temperature MR continued to progress gradually owing to the low MCE and high cost of Gd [14]. A significant breakthrough was made in 1997 when Gschneidner and Pecharchy discovered the giant MCE in the $Gd_5(Si_2Ge_2)$ alloy [15], which had twice the MCE of Gd [16]. Since then, extensive research in room-temperature MR has been conducted around the world, and there has been an exponential increase in relevant patents and papers [17]. Along with this research wave, numerous room-temperature MR prototypes have been established. According to the statistics, 41 room-temperature MR prototypes were built and tested before 2010 [18]. By the end of 2020, over 70 room-temperature MR prototypes were built. The details of these MR prototypes can be found in the reviews written by Yu et al. [18] and Greco et al. [19].

As reported by Greco et al. [19], the existing room-temperature MR prototypes can produce a maximum temperature span of 42 K under no load condition [20], or a maximum cooling capacity of 3024 W at zero temperature span [6]. However, a 2–10 kW cooling capacity is required over a 40 K temperature span in terms of the domestic air conditioner, and a 40 K temperature span at a hundred-watt cooling capacity is also necessary for the fresh maintenance of food and medicine [21]. Hence, room-temperature MR technology is currently unable to satisfy the refrigeration and cooling requirements in everyday life and is not commercially available. The direct reason for this is the inadequate temperature span and cooling capacity. Furthermore, the obstacles to achieving a sufficiently high temperature span and cooling capacity are as follows:

- ***Limited adiabatic temperature change of MCMs***

The MCM is the heart of an MR prototype. The greater the adiabatic temperature change of the MCM, the greater the temperature span of the prototype. Gd was adopted as the MCM by most of the previously described MR prototypes [22]. Other promising MCMs are Gd-based [23], LaFeSi-based [24], and Mn-based [25] alloys. The adiabatic temperature changes of these promising MCMs are summarized in Table 1. Currently, MCMs with greater adiabatic temperature changes are under active development. Regarding research on MCMs, many reviews have been published [36,37], and the most recent review can be found in the work of Zarkevich et al. [38]. The MCM-related progress is not addressed in this review.

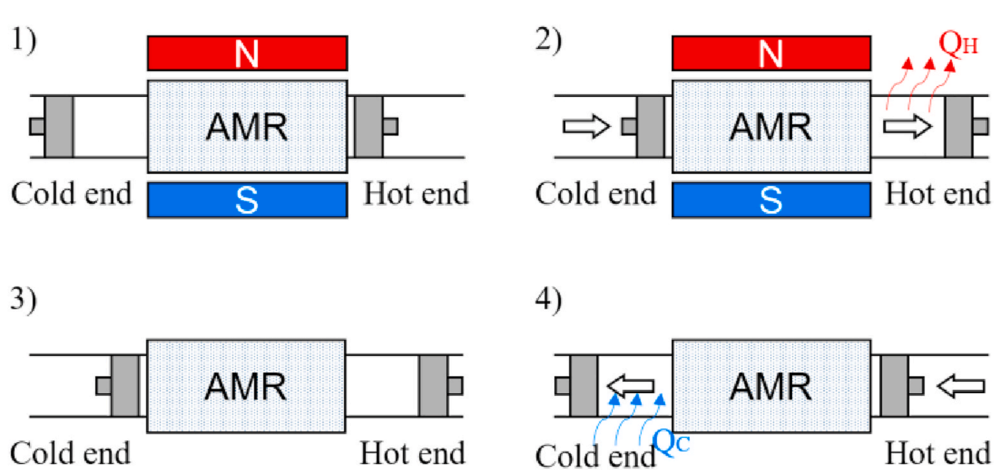



**Fig. 1.** Operating process of an AMR.

**Table 1**
Adiabatic temperature changes of various MCMs.

| References | MCMs | Curie temperature/K | Adiabatic temperature change/K | Magnetic field change/T |
|---|---|---|---|---|
| [26] | Gd | 292 | 4.7 | 2 |
| [27] | $Gd_{0.9}Tb_{0.1}$ | 286 | 1.9 | 1 |
| [15] | $Gd_5Si_2Ge_2$ | 278 | 7.3 | 2 |
| [28] | $LaFe_{11.2}Co_{0.7}Si_{1.1}$ | 274 | 2.4 | 2 |
| [29] | $LaFe_{11.06}Co_{0.86}Si_{1.08}$ | 276 | 2.3 | 1 |
| [29] | $LaFe_{11.05}Co_{0.94}Si_{1.01}$ | 287 | 2.1 | 1 |
| [29] | $LaFe_{10.96}Co_{0.97}Si_{1.07}$ | 289 | 2.2 | 1 |
| [30] | $La(Fe_{0.88}Si_{0.12})_{13}H$ | 274 | 6.2 | 2 |
| [30] | $La(Fe_{0.89}Si_{0.11})_{13}H_{1.3}$ | 291 | 6.9 | 2 |
| [30] | $La(Fe_{0.88}Si_{0.12})_{13}H_{1.5}$ | 323 | 6.8 | 2 |
| [31] | MnAs | 318 | 4.7 | 2 |
| [32] | $Mn_{1.1}Fe_{0.9}P_{0.47}As_{0.53}$ | 292 | 2.8 | 1 |
| [33] | $Ni_{55.4}Mn_{20}Ga_{24.6}$ | 313 | 1.5 | 2 |
| [34] | $Ni_{50.2}Mn_{35.0}In_{14.8}$ | 301 | 1.2 | 2 |
| [34] | $Ni_{49.6}Mn_{35.6}In_{14.8}$ | 268 | 1.6 | 2 |
| [35] | $Ni_{45.7}Mn_{36.6}Co_{4.2}In_{13.5}$ | 289 | 3.0 | 2 |

- ***Limited magnetic field intensity of permanent magnets***

In addition to the MCM, the magnet is another crucial component of an MR prototype. Magnets are of three main types: permanent magnets, superconducting magnets, and electromagnets. Earlier research on room-temperature MR used superconducting magnets, such as the first prototype built by Brown [9]. Currently, permanent magnets are widely used to provide magnetic fields for room-temperature MR prototypes. The reason permanent magnets are preferred is that they do not require power to generate a magnetic field [45]. Table 2 shows the common types of permanent magnets and their magnetic field intensities. From Table 2, the magnetic field intensity is lower than 2 T; therefore, permanent magnets with stronger magnetic field intensity are expected. Reviews on permanent magnets can be found in Refs. [45,46], and they are not a topic of this review.

- ***Theoretical limit of MR thermodynamic cycles***

The commonly used MR thermodynamic cycles include the magnetic Brayton and Ericsson cycles. These two cycles have their own advantages and limitations. Fig. 2 indicates that the magnetic Brayton cycle is characterized by a higher temperature span and lower cooling capacity compared with the magnetic Ericsson cycle. To simultaneously achieve a higher temperature span and higher cooling capacity, the current solution is to build a hybrid MR cycle, thereby combining the advantages of different MR cycles.

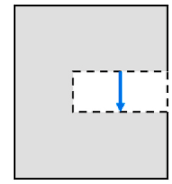
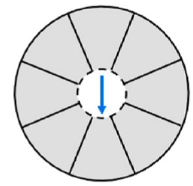
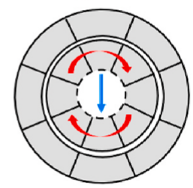
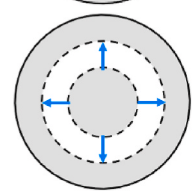
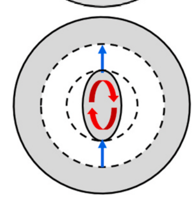
- ***Low operating frequency***

The operating frequency is defined as the number of MR thermodynamic cycles per unit of time. The low operating frequency directly leads to a decrease in the cooling capacity because the MR prototype outputs cooling capacity once per thermodynamic cycle. One of the major reasons for the low operating frequency is the limited convection heat transfer rate between the MCM and HTF [47], which results in the HTF being unable to absorb the heat/cooling capacity of MCM during a short period. Therefore, a long cycle period is necessary to ensure the heat transfer effect between the MCM and HTF. To shorten the cycle period and increase the operating frequency, two solutions are provided by current studies. One solution is to develop fully solid-state MR cycle independent of convection heat transfer, and the other is to enhance the heat transfer between the MCM and HTF.

- ***Other obstacles***

Other obstacles primarily consist of some irreversible loss, such as the spontaneous heat conduction inside the AMR from the hot to cold end, and the presence of flowing dead volume in the AMR. Current solutions reduce the irreversible loss from the point of heat transfer enhancement, flow path optimization, etc.

**Table 2**
Different types of permanent magnets and their magnetic field intensities.

| References | Shapes | Schematic diagram | Magnetic field intensity |
|---|---|---|---|
| [39,40,54,66] | C-shape |  | 1.9T [39,40]<br>1.5T [54]<br>0.875T [66] |
| [20,41,56] | Cylinder |  | 1.5T [20]<br>1.4T [41]<br>1.48–1.56T [56] |
| [57,58] | Nested Halbach cylinders |  | 1.3T [57]<br>1.4T [58] |
| [42,43] | Concentric Halbach cylinders |  | 0.98T [42]<br>1.24T [43] |
| [44,65] | Rotor-stator |  | 0.77T [44]<br>0.892T [65] |

In terms of inadequate temperature span and cooling capacity of room-temperature MR, in addition to seeking solutions for the obstacles mentioned above, another direct and efficient solution is combining MCE with other cooling effects (e.g. the gas regenerative cooling effect and caloric effect), and building an MR cycle combined with other cooling effects. Better cooling performance is expected to be achieved by a combination of multiple cooling effects.

In conclusion, the solutions stated above primarily address two areas. the first is MR thermodynamic cycles including the hybrid MR cycle, fully solid-state MR cycle and, MR cycle combined with other cooling effects. The other aspect is heat transfer enhancement during heat regeneration. Therefore, the solutions to obstacles in the commercialization of room-temperature MR are reviewed in this paper from two perspectives: MR thermodynamic cycle and heat transfer enhancement during heat regeneration. Moreover, this review outlines the application research of room-temperature MR, and provides the reader with a better understanding of its possible applications. Furthermore, some recommendations for future research are presented.

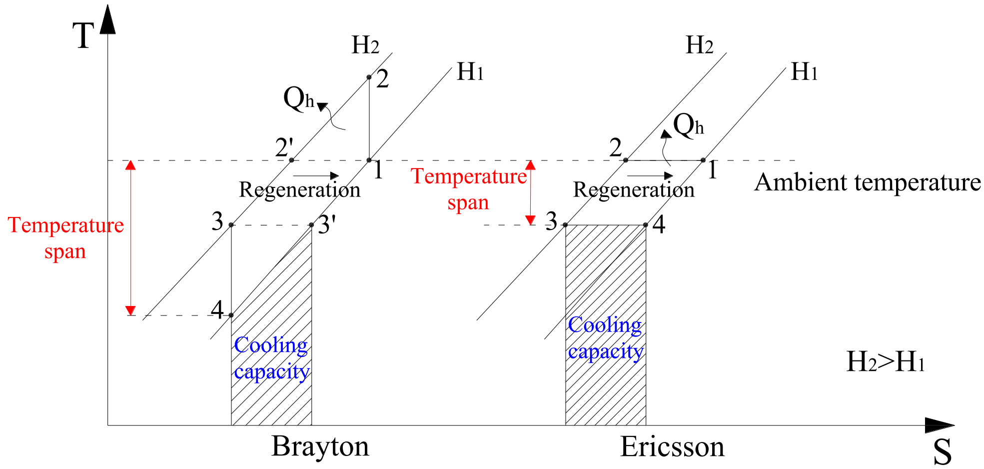



**Fig. 2.** Theoretical temperature span and cooling capacity achieved by the magnetic Brayton and Ericsson cycles.

## 2. MR thermodynamic cycle

### 2.1. Basic MR cycles and their hybrid cycles

Basic MR cycles primarily consist of a magnetic Carnot cycle, a magnetic Brayton cycle, and a magnetic Ericsson cycle [48,49]. As shown in Fig. 3a, the magnetic Carnot cycle consists of two adiabatic and two isothermal processes. It is usually used as a theoretical reference for performance comparisons between other MR cycles. The magnetic Brayton cycle (Fig. 3b) consists of two adiabatic and two iso-field processes. Compared with other basic MR cycles, the magnetic Brayton cycle is expected to achieve the largest temperature span. The magnetic Ericsson cycle (Fig. 3c) is similar to the magnetic Brayton cycle. The only difference is the isothermal (de)magnetization instead of adiabatic. In theory, when magnetic refrigerators operate in the magnetic Ericsson cycle, the maximum cooling capacity can be obtained. However, note that isothermal (de)magnetization is difficult to achieve.

To perform a basic MR cycle, a magnetic field variation matching this cycle is required. Fig. 4a displays an example of the magnetic field variation. This magnetic field variation is a sinusoidal function, and there is no region of constant field. Such a magnetic field variation can be applied to the magnetic Carnot cycle. Another example of magnetic field variation is shown in Fig. 4b. It involves only one constant magnetic field. Such a magnetic field variation can serve for the magnetic Brayton and Ericsson cycles. In addition, a constant magnetic field is usually provided by a permanent magnet. To create the largest possible MCE for the MCM, the permanent magnet should produce a region with a high magnetic flux density preferably with as high homogenization as possible [45]. However, in most scenarios, the homogenization of the magnetic flux distribution results in complex permanent magnet structures and a decrease in the maximum magnetic field [50,51].

Over recent years, much interest has been given to the hybrid MR cycle combining different basic MR cycles. In the study of Plaznik et al. [52], Brayton, Ericsson, and hybrid Brayton-Ericsson cycles (Fig. 5) with an AMR were compared based on numerical and experimental analyses. Based on numerical simulations, Plaznik et al. [52] concluded that a higher coefficient of performance (COP) and a higher cooling capacity could be achieved with hybrid MR cycle compared with the Brayton and Ericsson cycles, respectively. Based on the experimental results, the maximum temperature span of hybrid MR cycle was more than 10% higher than that of the Brayton and Ericsson cycles. As a result, they reported that the hybrid MR cycle exhibited a better overall performance.

As reported by Plaznik et al. [52], the operation of the above three MR cycles with an AMR relied on the control of the fluid flow period (Fig. 6). As Fig. 6 shows, with the same magnetic field variation, the change in the fluid flow period affected both the type of MR cycles and proportion of different basic MR cycles in the hybrid MR cycle. Except for the hybrid MR cycle shown in Fig. 5, Kitanovski et al. [53] proposed other potential hybrid MR cycles (Fig. 7). A comparison of Figs. 5 and 7 indicates that all three hybrid MR cycles were a combination of the magnetic Brayton and Ericsson cycles, and the difference was the performing order of basic MR cycles. Therefore, for various requirements, the proportion and performing order of different basic MR cycles in the hybrid MR cycles are worth serious consideration in practical applications.

In addition to hybrid MR cycles composed of various basic MR cycles, multiple hybrid cycle modes based on a single basic MR cycle have also been of interest. He et al. [54] presented three hybrid cycle modes based on the magnetic Brayton cycle: series, parallel, and cascade cycle modes.

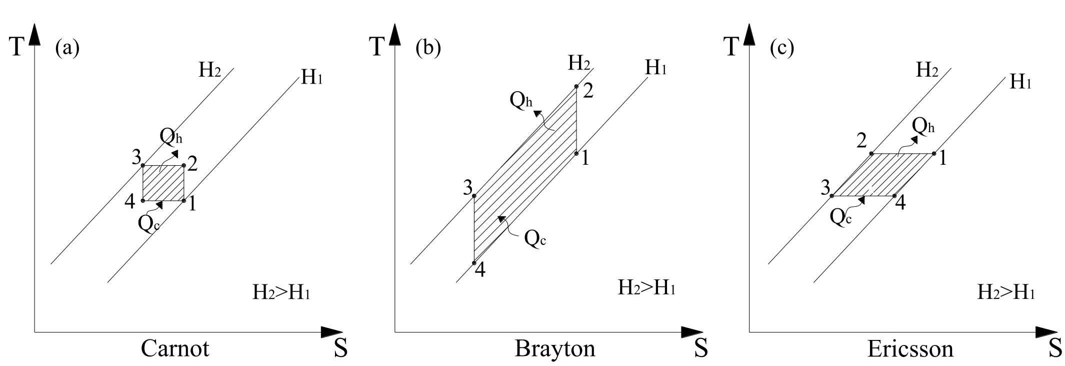



**Fig. 3.** Basic MR cycles: (a) magnetic Carnot cycle; (b) magnetic Brayton cycle; (c) magnetic Ericsson cycle.

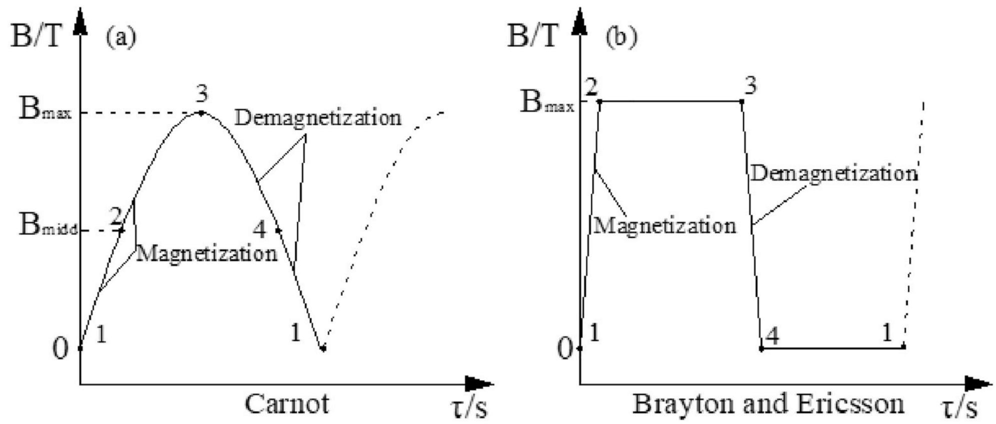



**Fig. 4.** Magnetic field variation of basic MR cycles: (a) magnetic Carnot cycle; (b) magnetic Brayton cycle and magnetic Ericsson cycle.

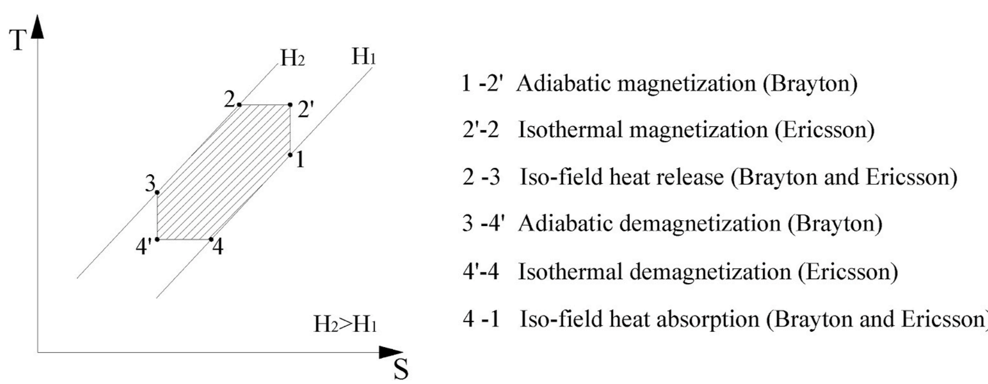



**Fig. 5.** Hybrid Brayton-Ericsson cycle.

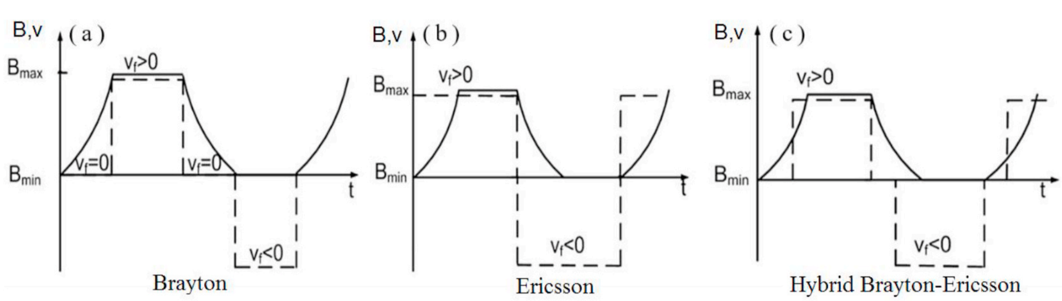



**Fig. 6.** Flowing regimes for three thermodynamic cycles with an AMR: (a) magnetic Brayton cycle; (b) magnetic Ericsson cycle; (c) hybrid Brayton-Ericsson cycle [52].

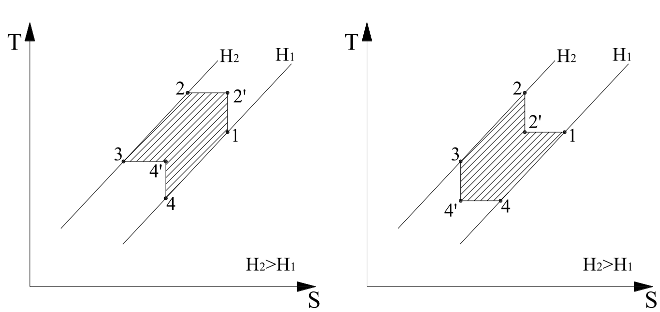



**Fig. 7.** Other potential hybrid MR cycles.

Generally, the series and parallel cycle modes are more conducive to obtaining a larger temperature span and higher specific cooling power (SCP), respectively (Fig. 8). The cascade cycle mode is expected to combine the advantages of both the series and parallel cycle modes. To compare the cooling performance of three hybrid cycle modes, an MR system was designed by He et al. [54], as shown in Fig. 9. The principles and characteristics of three hybrid cycle modes are listed in Table 3. The comparative results indicated that the maximum temperature span and maximum SCP of the cascade cycle mode were higher than those of the series cycle mode. Moreover, a higher maximum SCP could be achieved with cascade cycle mode compared with the parallel cycle mode at a larger temperature span. Therefore, He et al. [54] concluded that the cascade cycle mode has good prospects for application, but the problem of heat transfer matching between the high- and low-temperature stages

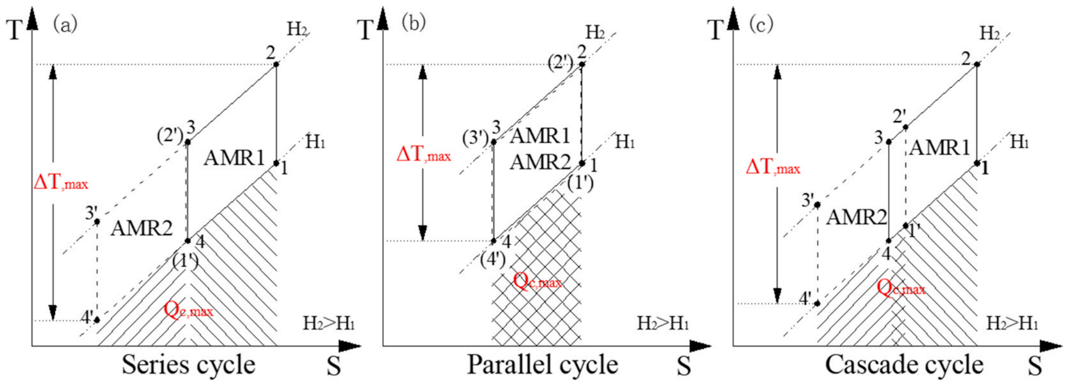



**Fig. 8.** T–S diagrams of three hybrid cycle modes: (a) series; (b) parallel; (c) cascade.

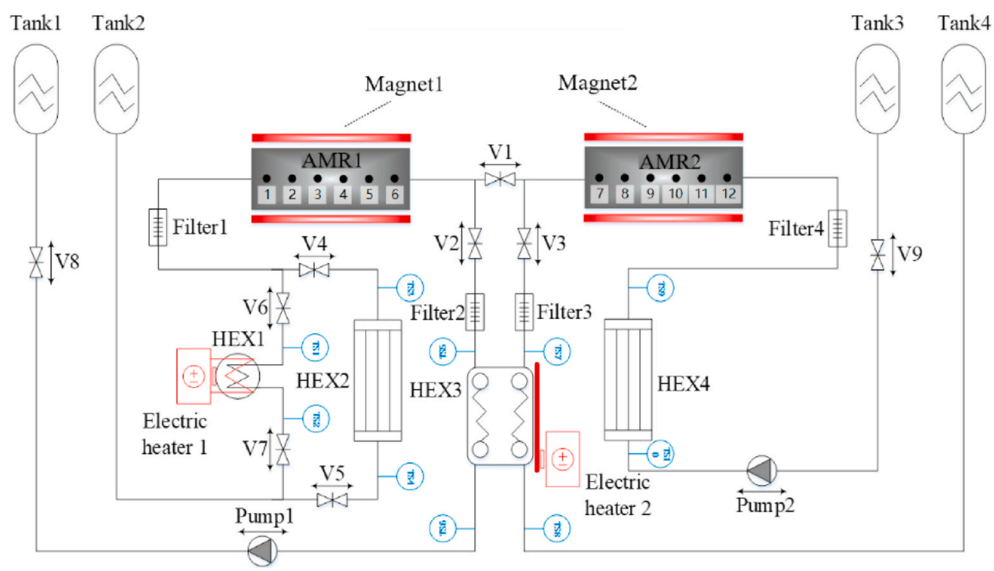



**Fig. 9.** MR system with three hybrid cycle modes [54].

**Table 3**
Operating principle and characteristics of three hybrid cycle modes.

| Modes | Status of valves | Characteristics of hybrid cycle modes |
|---|---|---|
| Series | On-state: 1, 6, 7, 9<br>Off-state: the rest | AMR1 and AMR2 are in the same fluid circuit;<br>AMR1 and AMR2 are (de)magnetized simultaneously. |
| Parallel | On-state: 2, 3, 4, 5, 8, 9<br>Off-state: the rest | AMR1 and AMR2 are in two different fluid circuits;<br>The plate heat exchanger (PHE) is utilized as the mutual CHEX of AMR1 and AMR2;<br>AMR1 and AMR2 are (de)magnetized simultaneously. |
| Cascade | On-state: 2, 3, 6, 7, 8, 9<br>Off-state: the rest | AMR1 and AMR2 are in two different fluid circuits;<br>The PHE is utilized as the intermediate heat exchanger between the low temperature stage (AMR1) and the high temperature stage (AMR2);<br>AMR1 and AMR2 are (de)magnetized alternately. |

must be solved.

Based on the above discussion, both the hybrid MR cycle composed of various basic MR cycles and the multiple hybrid cycle modes based on a single basic MR cycle have a positive effect on the cooling performance of magnetic refrigerators. Whereas, the proportion and performing order of different basic MR cycles in the hybrid MR cycles should be considered in practical applications. Besides, inter-stage heat transfer matching in the cascade cycle mode should be considered.

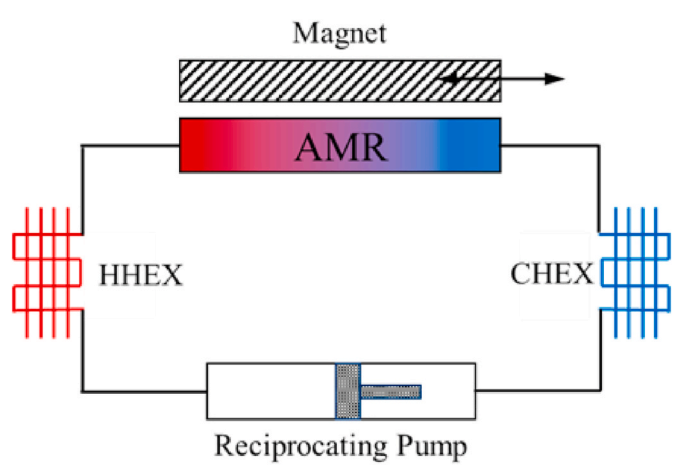



**Fig. 10.** Schematic diagram of the classic cycle.

### 2.2. Optimal design of the AMR cycle

The design of a classic AMR cycle is displayed in Fig. 10. This classic cycle is made up of five parts: the magnet, AMR, hot end heat exchanger (HHEX), cold end heat exchanger (CHEX), and reciprocating pump. As descripted in Section 1, the classic cycle consists of four operating processes: magnetization, cold to hot flow, demagnetization, and hot to cold flow. In these four operating processes, note that no HTF flows through the MCM during the magnetization and demagnetization processes. Such an operating cycle is defined as a discontinuous cycle [20], which causes a decrease in SCP. Besides, the HTF flows back and forth through the MCM after each magnetization and demagnetization, which is considered as an oscillating cycle. This oscillating feature leads to an increase in the irreversible loss owing to the existing flowing dead volume [55].

As an answer to the discontinuous cycle and oscillating cycle, a continuous cycle [20,56] and a unidirectional cycle [57] have been designed by some researchers, respectively.

The continuous cycle was utilized in the MR prototypes developed by Tura et al. [58,59], Velázquez et al. [60], Lee [61], and many other researchers [62–64]. Fig. 11 exhibits the schematic diagram of the continuous cycle. Unlike the classic cycle, the continuous cycle consists of two AMRs and two HHEXs. The operating process of continuous cycle can be described as follows:

1) In the initial state, AMR1 stays inside the magnetic field, but AMR2 stays outside the magnetic field.
2) The magnet moves rightward, so AMR1 is demagnetized while AMR2 is magnetized. Simultaneously, the HTF is pumped from HHEX1 to HHEX2. During flow, the HTF is cooled down in AMR1 and then absorbs heat from CHEX, after which the HTF is heated in AMR2 and finally releases heat to HHEX2.
3) The magnet moves leftward, and AMR1 is magnetized, and AMR2 is demagnetized. Simultaneously, the HTF flows from HHEX2 to HHEX1. The HTF is cooled down in AMR2 and then absorbs heat from CHEX; subsequently, the HTF is heated in AMR1 and finally releases heat to HHEX1.

By repeating the above processes, a continuous cooling output can be achieved.

The operating process of the continuous cycle shows that the HTF flows through the AMR simultaneously when the AMR is (de)magnetized, which indicates that the HTF flow is continuous. Due to the continuous flow of the HTF, the magnetic refrigerator using a continuous cycle can output cooling capacity every operating process. However, for the classic cycle, the magnetic refrigerator outputs the cooling capacity once every four operating processes. Thus, an improvement in the cooling capacity output is most likely to be achieved by the design of continuous cycle.

The unidirectional cycle was utilized in the MR prototypes developed by Gao et al. [13] and Li et al. [57]. Fig. 12 depicts the schematic diagram of the unidirectional cycle, which adopts the design of double fluid circuits instead of the single fluid circuit of the classic cycle. The operating process of the unidirectional cycle can be described as follows:

1) The MCM in the AMR is magnetized and its temperature increases.

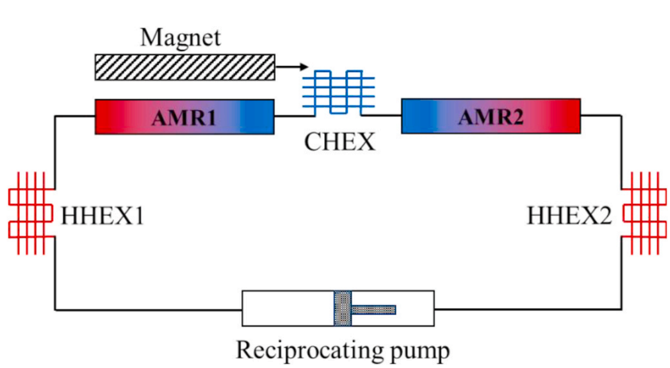



**Fig. 11.** Schematic diagram of the continuous cycle.

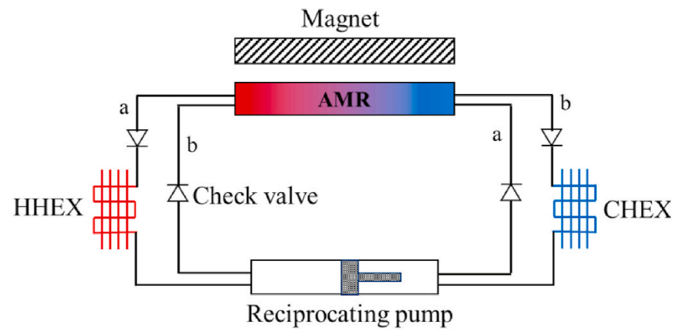



**Fig. 12.** Schematic diagram of the unidirectional cycle.

2) After absorbing heat from CHEX, the HTF flows through circuit a. Then, the HTF is heated by the MCM in the AMR and releases heat to the HHEX.
3) The MCM is demagnetized and its temperature decreases.
4) The HTF, after releasing heat to the HHEX, flows through circuit b. In the flowing process, the HTF is cooled down by the MCM, and then absorbs heat from the CHEX.

The description of the unidirectional cycle described above indicates that after the magnetization and demagnetization, the HTF flows through circuits a and b, respectively. Hence, the flow of the HTF outside the AMR is unidirectional. This design of double fluid circuits for unidirectional cycle is conducive to reducing the dead volume, and further reducing the irreversible loss during heat regeneration.

As a combination of the continuous cycle and unidirectional cycle, the cycle shown in Fig. 13 has been much more popular over the last five years. The operating process of this cycle is as follows:

1) AMR1 is magnetized and AMR2 is demagnetized. The HTF flows along the black arrows. During flow, the HTF is heated in AMR1 and then releases heat to the HHEX, after which the HTF is cooled down in AMR2 and absorbs heat from the CHEX.
2) AMR1 is demagnetized and AMR2 is magnetized. The HTF flows along the red arrows. The HTF is heated in AMR2 and releases heat to the HHEX; subsequently, the HTF is cooled down in AMR1 and finally absorbs heat from the CHEX.

As described above, such a cycle can output cooling capacity in each process, which is in accordance with the characteristic of continuous cycle. Besides, the flow of the HTF outside the AMR is unidirectional, which is in accordance with the characteristic of unidirectional cycle. Therefore, this cycle has the advantages of high cooling capacity and low irreversible loss. MR prototypes using this cycle were designed by Lozano et al. [65], Huang et al. [66] and Nakashima et al. [67].

In summary, the optimal design for the fluid circuit of the classic cycle may be a good option to increase the cooling capacity and reduce the irreversible loss during heat regeneration. However, the complex fluid circuit introduces a more complex system and more difficult flow control mechanism.

### 2.3. Fully solid-state MR cycle

As a promising solution to the low operating frequency, the fully solid-state MR cycle, using thermal switches to replace the HTF, has been the subject of numerous MR studies. The thermal switch, as a key component of fully solid-state MR system, is utilized to manipulate and control the heat transfer direction and occasionally the heat transfer rate [68]. Depending on whether an external power supply is required, the thermal switch can be divided into active and passive thermal switches.

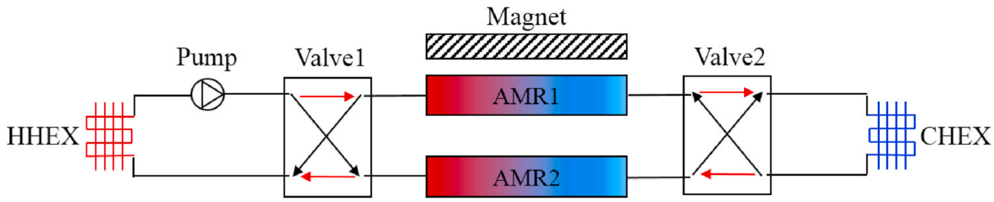



**Fig. 13.** Cycle combining the features of continuous and unidirectional cycle.

*2.3.1. Fully solid-state MR cycle based on active thermal switch*

The active thermal switch primarily refers to the Peltier element. The basic structure of a fully solid-state MR model based on Peltier elements is a sandwiched structure, which is a thin layer of MCM sandwiched between two layers of Peltier elements. By controlling the on-off operation of Peltier elements, heat can be transferred quickly between the MCM and HHEX or CHEX.

Tomc et al. [68] developed a single-stage model of the fully solid-state MR cycle based on Peltier elements, as shown in Fig. 14. When the MCM was magnetized, Peltier 1 was turned on and heat was transferred quickly from the MCM to the HHEX. Peltier 2 was turned off and functioned as a thermal insulator between the MCM and CHEX. When the MCM was demagnetized, Peltier 2 was turned on, and the MCM absorbed heat quickly from the CHEX. Peltier 1 was turned off and functioned as a thermal insulator between the MCM and HHEX. To evaluate the operating characteristics of such a single-stage MR device, a quasi-2D numerical model was built. The simulation results showed that this single-stage MR device could operate at a rather high operating frequency of over 200 Hz. Besides, they concluded that the key factors affecting the operating frequency were the heat flux rate of Peltier, the MCM thickness, and the temperature span. Among them, the heat flux rate of Peltier was the most significant. An increase in the heat flux rate of Peltier increased the operating frequency, and subsequently enhanced the SCP. However, the higher the heat flux rate of Peltier, the higher the electric current that was required, which resulted in an increase in Joule heating and a decrease in exergy efficiency (the exergy efficiency decreased from 65% to 10% as the heat flux rate of Peltier increased from 0.041 to 0.327 W).

Following the research of Tomc et al. [68], a more elaborate single-stage model employing the same Peltier element (type MPC-D701) was established by de Vries et al. [69], as shown in Fig. 15a. In this model, the resistance of the Peltier element was obtained with higher accuracy by modelling the full geometry. Moreover, unlike the two fluid circuits (Fig. 14b) utilized by Tomc et al. [68], heat regeneration was achieved by employing a single fluid circuit and routing the CHEX outlet to the HHEX inlet, as displayed in Fig. 15b. Contrary to the results of Tomc et al. [68], de Vries et al. [69] considered that the application outlook of Peltier elements was less optimistic. In addition, they concluded that the single-stage MR device based on MCE and Peltier effect exhibited a worse performance than that based only on the Peltier effect because of the heat leakage through the off-state Peltier element.

In addition to commercially available Peltier elements, Egolf et al. [70] applied a Ni-nanowire Peltier element to the single-stage model. The Ni-nanowire Peltier element was characterized by its ultra-thinness and flexibility, which made it suitable for coating almost any surface. Nonetheless, several important parameters should be further optimized in future practical applications, such as electric resistance, thermal resistance, and maximal switching frequency. Moreover, some practical aspects should be investigated, such as the durability and cost of the Ni-nanowire Peltier element.

Along with the single-stage model, the cascade model is also being researched. The concept of a cascade model was first introduced by Tasaki et al. [71] for in-vehicle applications. Subsequently, based on the work of Tasaki et al. [71], an extended study was conducted by Olsen et al. [72]. Later, Monfared et al. [73] built a more detailed cascade model of the fully solid-state MR cycle based on Peltier elements, visible in Fig. 16. This cascade model included several layers of MCMs and Peltier elements. The MCM was sandwiched between two Peltier elements. Refrigeration was achieved through a unidirectional heat transfer controlled by Peltier elements in conjunction with the temperature variation of the MCMs through MCE. According to the simulation for this cascade model, Monfared et al. [73] indicated that increasing the electric current and length of Peltier elements eventually decreased the COP and SCP owing to Joule heating. Thus, the electric current and geometry of the Peltier elements should be optimized to achieve high performance. In addition, more efficient thermoelectric semiconductor materials are expected to minimize Joule heating.

In the above cascade model, heat was transferred solely via heat conduction, and no active regeneration occurred. Meanwhile, to achieve unidirectional heat transfer from the cold end to hot end, Peltier elements were switched frequently between the on and off states. To overcome these limitations, Wu and Lu [74,75] designed a novel fully solid-state MR model based on a new cascade micro-unit regeneration (MUR) cycle, as shown in Fig. 17a. The main concept of the MUR cycle was each higher temperature MCM lattice remaining inside the magnetic field and the paired lower temperature MCM lattice remaining outside the magnetic field exchanged heat with each other automatically through heat transfer blocks (Fig. 17b), thereby realizing heat regeneration and enlarging the temperature span. The specific structure and operating principle of this novel model were described in detail by Wu [74]. Furthermore, the effects of three heat transfer configurations

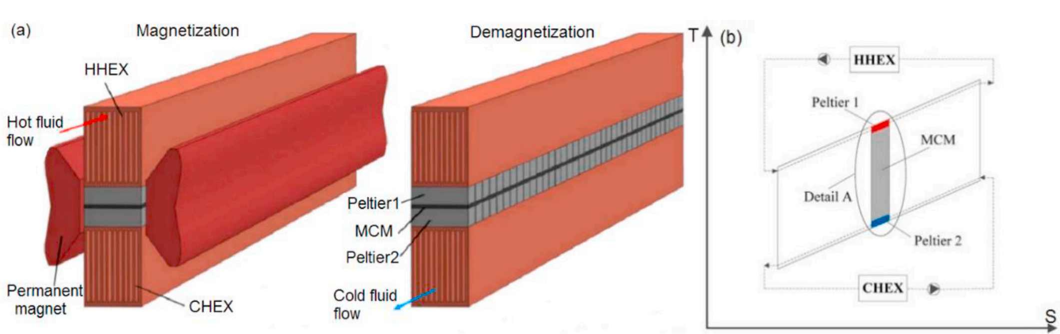



**Fig. 14.** Single-stage model of the fully solid-state MR cycle based on Peltier elements: (a) geometry; (b) T-S diagram [68].

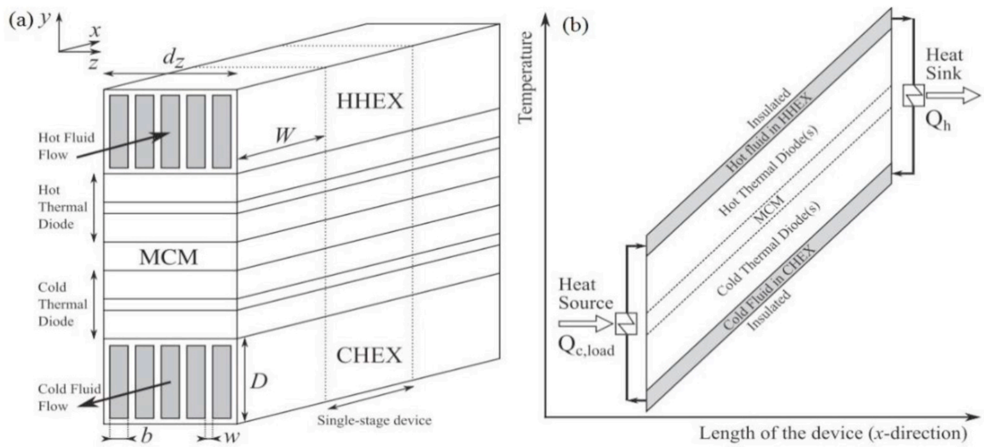



**Fig. 15.** Another single-stage model of the fully solid-state MR cycle based on Peltier elements: (a) geometry; (b) temperature distribution over the length of the MR device [69].

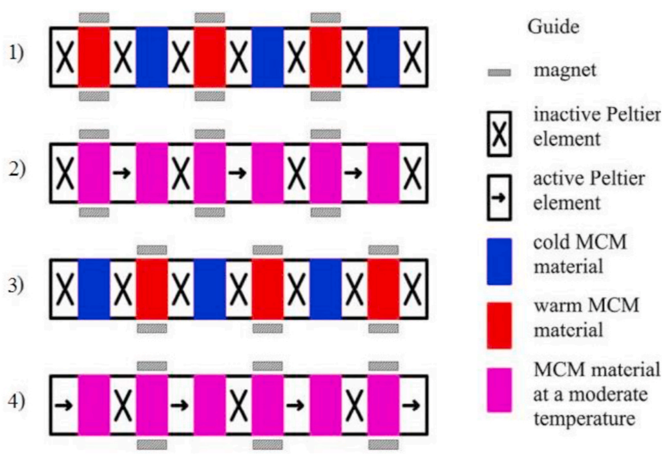



**Fig. 16.** Cascade model of the fully solid-state MR cycle based on Peltier elements [73].

(Fig. 18) and two middle heat transfer blocks (copper blocks and Peltier elements) on the cooling performance were investigated. The results revealed that when using copper blocks for heat regeneration, the maximum SCP obtained by the topology optimization structure was 32% and 391% higher than that of the parallel sheet and Gd-only structures, respectively. In addition, when using Peltier elements for fast heat regeneration, the maximum SCP increased by 82%–149% compared with the heat regeneration using copper blocks. Thus, the cooling performance of the MUR cycle was significantly improved by using Peltier elements and topology optimization structure composed of copper and Gd.

To this point, the cooling performance of the fully solid-state MR cycle based on Peltier elements has been discussed, and the results indicate that significant improvements in the operating frequency and SCP can be achieved when using the Peltier element as the thermal switch. However, Peltier elements have two major limitations for practical applications, namely Joule heating and heat leakage. Therefore, the operating parameters and structural parameters of Peltier elements should be optimized in future studies to minimize Joule heating and heat leakage. In addition, a novel Peltier element fabricated from more efficient thermoelectric materials is required.

### 2.3.2. Fully solid-state MR cycle based on passive thermal switch

The passive thermal switch contains $k_H$ material, low thermal conductivity material (LTCM) and high thermal conductivity material (HTCM). The $k_H$ material is a material whose thermal conductivity k depends on the applied magnetic field H [76].

In recent years, a series of studies on the fully solid-state MR cycle based on $k_H$ materials have been performed by Silva et al. from the University of Porto [76–79]. In 2012, Silva et al. [76] first developed a single-stage model of the fully solid-state MR cycle based on $k_H$ materials, as displayed in Fig. 19. When no magnetic field was applied (H = 0), $k_{1(0)}<k_{2(0)}$, and heat was easily transferred from the cold reservoir to the MCM. After applying a magnetic field (H > 0), $k_{1(0)}+\Delta k_1>k_{2(0)}-\Delta k_2$, heat was easily transferred from the MCM to the hot reservoir. With the assumption of $k_1 = 0$ and $k_2 = 10.5$ W/(m K) for H = 0, and $k_1 = 10.5$ W/(m K) and $k_2 = 0$ for H = 1 T, a 1D numerical simulation was conducted to test the viability of this single-stage model. The simulation results indicated that a maximum SCP of 2.75 W/cm$^2$ could be obtained at an operating temperature of 296 K, operating frequency of 95 Hz, and

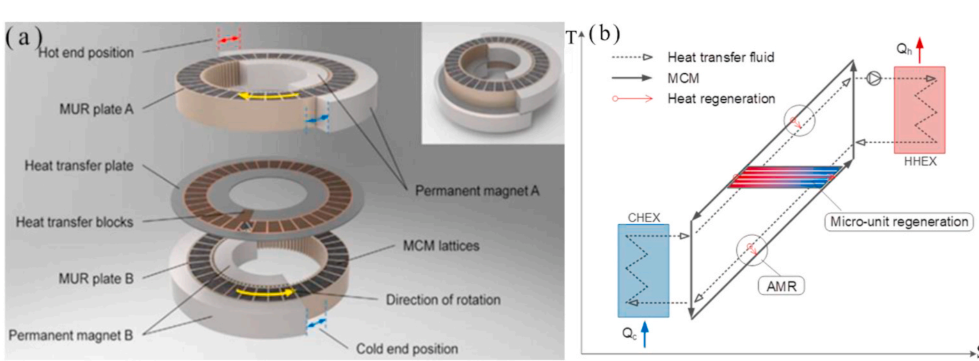



**Fig. 17.** Fully solid-state MR model based on MUR cycle: (a) MR system; (b) T-S diagram of MUR cycle [74].

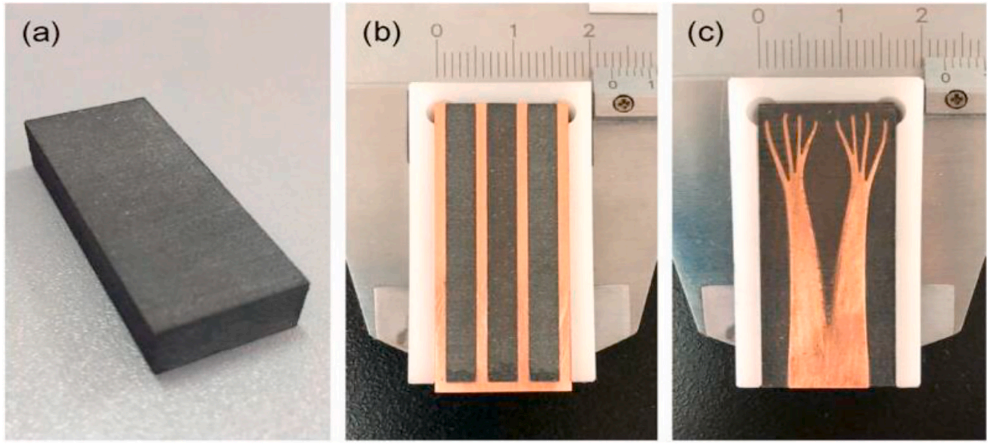



**Fig. 18.** Image of three heat transfer configurations (a) Gd only; (b) parallel sheets; (c) topology optimization structure [75].

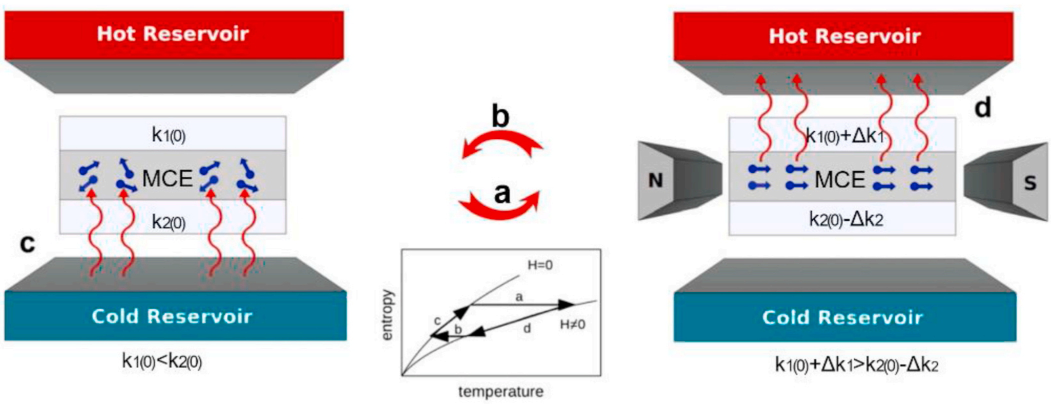



**Fig. 19.** Single-stage model of the fully solid-state MR cycle based on $k_H$ materials [76].

Gd thickness of 1 mm. Moreover, the maximum temperature span of this single-stage model was limited by the adiabatic temperature variation of the MCM, and the obtained maximum temperature span was 2.5 K, which was too small to satisfy the requirements of practical cooling.

To increase the temperature span, a cascade model of the fully solid-state MR cycle based on $k_H$ materials was proposed and simulated by Silva et al. in 2014 [77], as shown in Fig. 20. In this cascade model, the MCM was divided into several units. First, a magnetic field was applied to the entire MCM. Here, $k_{H1}$ was active and $k_{H2}$ was inactive; thus, heat was transferred from the MCM to the hot reservoir. Subsequently, the magnetic field was removed from the hot reservoir to the cold reservoir. When the magnetic field was removed from the final MCM unit, $k_{H1}$ was

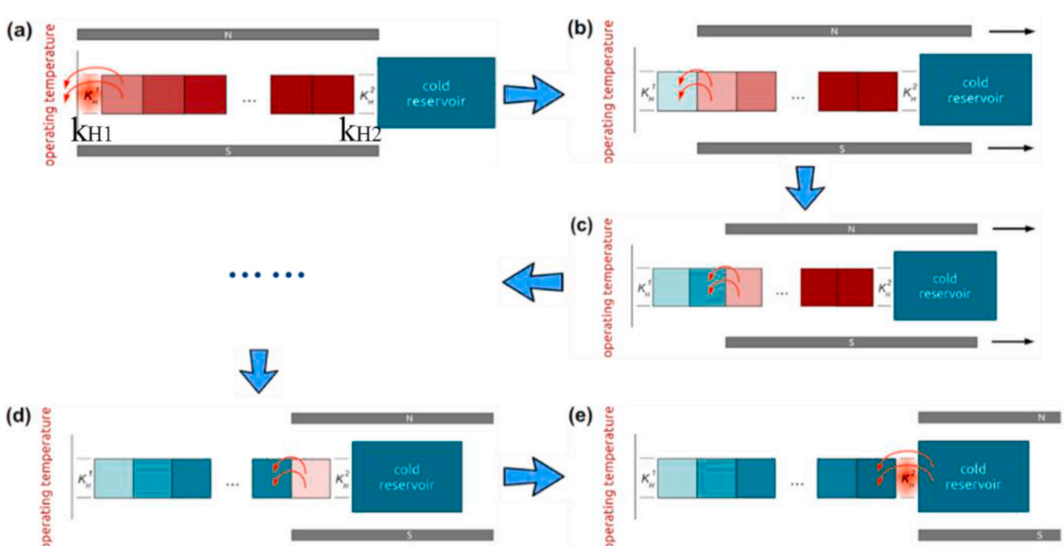



**Fig. 20.** Cascade model of the fully solid-state MR cycle based on $k_H$ materials [77].

inactive and $k_{H2}$ was active, and the last MCM unit absorbed heat from the cold reservoir. According to the numerical simulation, Silva et al. [77] reported that a maximum temperature span of 11.5 K could be achieved, which was 450% higher than that of the MCM unit.

Based on the study of the cascade model [77], the effects of MCM physical properties [78] and magnetic field operating modes [79] on the temperature span were further investigated by Silva et al. The results indicated that the temperature span was proportional to the square root of the density, heat capacity and adiabatic temperature variation of the MCM, but inversely correlated with its thermal conductivity. In addition, the largest temperature span was obtained when the magnetic field was removed from the hot reservoir to the cold reservoir in an accelerated motion.

In summary, the concept of using $k_H$ materials as thermal switches in the fully solid-state MR model demonstrates an optimistic prospect, but $k_H$ materials are rather difficult to attain. Currently, candidate materials are materials with a giant magnetoresistive effect, magnetic nanofluids, and spintronic materials [76].

For LTCMs, Utaka et al. [80] proposed a simple and effective thermal switch (Fig. 21) for the cascade model (Fig. 22). This thermal switch function was achieved by the relative rotation of the adjacent MCM units. Specifically, the interface between two adjacent MCM units was half-coated with a LTCM for thermal insulation. With the rotation of MCM units, heat was conducted between adjacent MCM units when surfaces with the same thermal conductivity overlapped, while MCM units were insulated when surfaces with different thermal conductivities overlapped. Based on the implementation mechanism of the thermal switch, Utaka et al. [80] concluded that when heat was conducted, the spontaneous heat conduction from the hot end to the cold end had a significantly negative effect on the COP and SCP. Reducing the spontaneous heat conduction was the key to obtaining a high cooling performance.

For HTCMs, Momen and Zhang et al. [81,82] designed a fully solid-state MR model consisting of an MCM body and reciprocating moving rods/sheets with HTCM (e.g. copper), as displayed in Fig. 23. The feature for this MR model was that the moving rods/sheets were longer than the MCM body, and the extended part behaved like heat exchangers. Specifically, when the MCM body was magnetized, the moving rods/sheets moved from the cold end to the hot end, and the extended part (behaving as a HHEX) released heat to the hot space. When the MCM body was demagnetized, the moving rods/sheets moved in the opposite direction, and the extended part (behaving as a CHEX) absorbed heat from the cold space. To identify the important factors affecting the cooling performance, a sensitive analysis was performed. The results revealed that effective surface roughness affected the COP and SCP the most.

In a word, LTCMs and HTCMs can be easily implemented in the fully solid-state MR, and the materials are much easier to obtain. However, the spontaneous heat conduction from high to low temperature cannot be ignored when applying LTCMs and HTCMs as thermal switches. Moreover, other factors that increase the irreversible loss require to be considered, such as the friction heat produced by the relative movement of components and air thermal resistance.

Thus far, we have introduced the existing fully solid-state MR cycle based on active and passive thermal switches. Compared with the active thermal switch, the passive thermal switch seems promising from the perspective of efficiency due to no power consumption nor Joule heating. However, the heat transfer rate of the passive thermal switch cannot be adjusted to satisfy the requirements of different operating conditions. More importantly, the function realization of passive thermal switches, particularly $k_H$ materials, is highly dependent on the material properties. Hence, the breakthrough in material science is the basis for applying the fully solid-state MR based on passive thermal switches.

The thermosiphon, which is also known as the gravity heat pipe, has the characteristic of bottom-up unidirectional heat transfer. Therefore, the gravity heat pipe is also a thermal switch that can be used to control the heat transfer direction [83]. The fully solid-state MR model based on gravity heat pipes is similar to that based on Peltier elements. The only difference is in the placement direction. That is, in the MR model based on gravity heat pipes, the HHEX, gravity heat pipe, MCM, another gravity heat pipe, and CHEX must be placed vertically in turn. To date, such a fully solid-state MR model has not been investigated in the scientific literature.

In conclusion, significant improvements in the operating frequency and cooling capacity can be achieved by introducing the fully solid-state MR cycle mechanism. However, all of the obtained results were based on numerical simulations. Therefore, fully solid-state MR prototypes should be built to experimentally verify the feasibility of fully solid-state MR cycle.

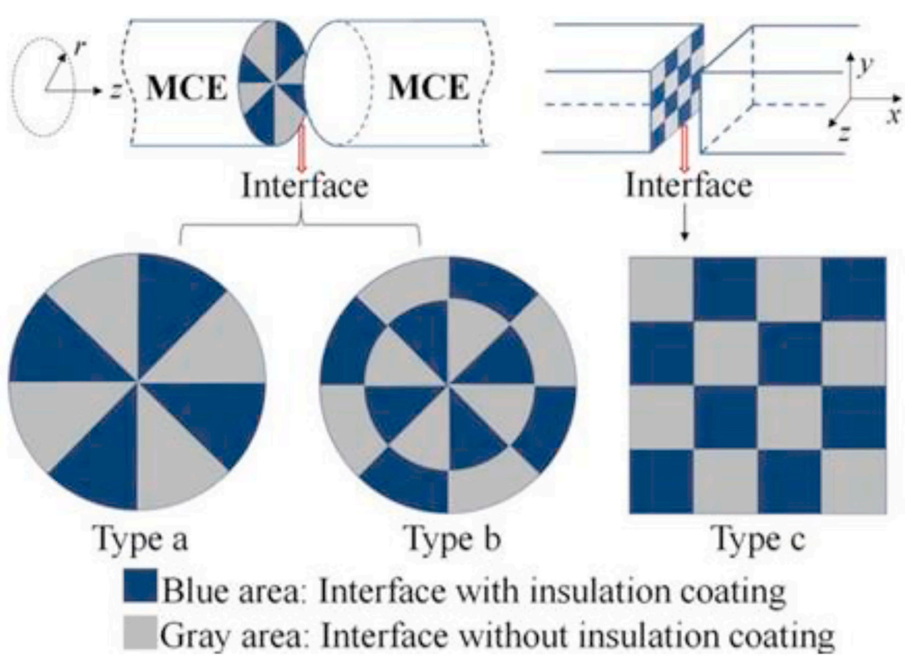



**Fig. 21.** Thermal switch with LTCM [80].

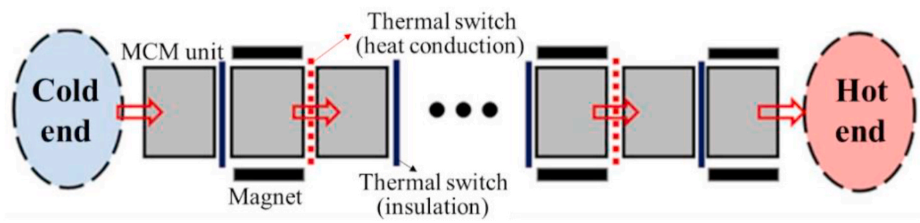



**Fig. 22.** Cascade model of the fully solid-state MR cycle based on LTCM [80].

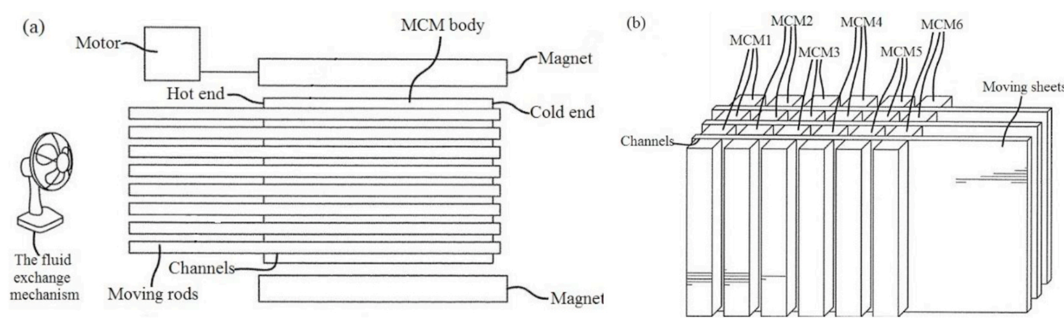



**Fig. 23.** Fully solid-state MR model with moving parts: (a) moving rods; (b) moving sheets [81].

### 2.4. MR cycle combined with other cooling effects

Most recently, the MR cycle combined with other cooling effects has become increasingly popular owing to its remarkable total cooling effect. Other cooling effects can be the gas regenerative cooling effect and caloric effects (e.g. electrocaloric effect, mechanocaloric effect, etc.).

#### 2.4.1. Hybrid cycle combining MCE and gas regenerative cooling effect

The MCE and gas regenerative cooling effect can be combined because the MR cycle (referring to the AMR cycle) and the gas regenerative refrigeration cycle are similar in heat transfer structure and flow path control. That is, both cycles include a regenerator and a device driving the HFT reciprocating flow.

Examples of combining the MR cycle and gas regenerative refrigeration cycle are mainly the hybrid cycle combining the MR cycle and GM cycle [84,85], and the hybrid cycle combining the MR cycle and Stirling cycle [86].

Fig. 24 displays the schematic diagram of the hybrid refrigerator combining the MR cycle and GM cycle. It can be seen from Fig. 24 the hybrid refrigerator and the GM refrigerator are similar. The difference between them is that the regenerator of the hybrid refrigerator is filled with MCMs, and a magnetic field is applied outside the regenerator.

The operating processes of this hybrid refrigerator that combines the MR cycle and GM cycle can be described as follows:

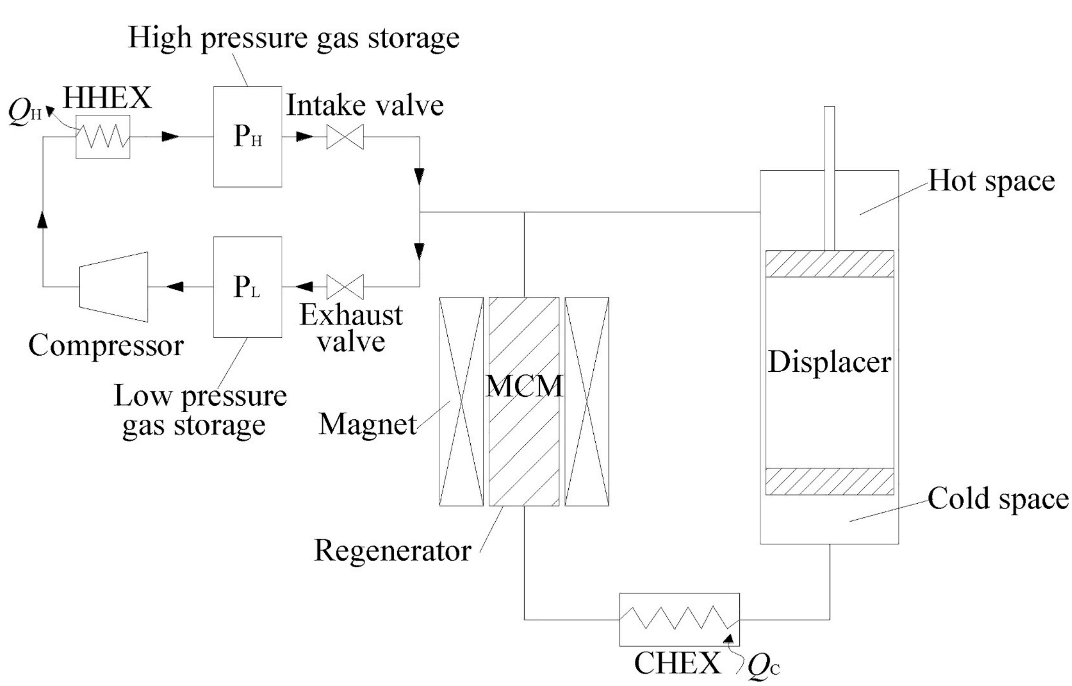



**Fig. 24.** Schematic diagram of the hybrid refrigerator combining the MR cycle and GM cycle.

1) Compression process: the helium gas is compressed by the compressor, and releases heat $Q_H$ to the HHEX. Subsequently, the displacer is at the bottom and is not moved. The intake valve is opened, and helium gas is introduced from the high-pressure gas storage to the hot space and regenerator.
2) Hot to cold blow/demagnetization process: the intake valve remains open. The displacer moves up to the topmost point, pushing the helium gas from the hot space through the regenerator to the cold space via an isobaric process. The helium gas is cooled down by the MCM in the regenerator as it passes through the regenerator. Meanwhile, the MCM is demagnetized and further absorbs heat from helium gas.
3) Expansion process: the displacer is at the top and is not moved. The intake valve is closed, and the exhaust valve is opened. The helium gas inside the cold space expands and absorbs heat $Q_C$ from CHEX, achieving refrigeration.
4) Cold to hot blow/magnetization process: the exhaust valve remains open, and the displacer moves down to the bottommost point. In this process, the helium gas is pushed in the opposite direction (relative to process 2) via an isobaric process. The helium gas absorbs heat from the MCM when it passes through the regenerator. Simultaneously, the MCM is magnetized, and rejects more heat to the helium gas.

Fig. 25 shows the T-S diagram of the hybrid cycle combining the MR cycle and GM cycle. When the helium gas is transferred from the hot space to the cold space through the regenerator (3–4 in the T-S diagram), the MCM is demagnetized. The demagnetization of the MCM enables it to absorb more heat from the helium gas. As a result, the helium gas is delivered to the cold space at a lower temperature compared with the process without MR. Similarly, when the helium gas is pushed from the cold space to the hot space through the regenerator (6-1 in T-S diagram), the MCM is magnetized. The magnetization of the MCM enables it to reject more heat to the helium gas. Consequently, compared with no MR, the helium gas is delivered to the hot space at a higher temperature. During these two processes of heat transfer between MCM and helium gas, the MCM serves as both a heat regeneration medium for the GM cycle and a refrigerant for the MR cycle. Thus, the MCM is the combining point between the GM cycle and MR cycle. Furthermore, as Fig. 25 shows, the MCM is demagnetized at a higher temperature while magnetized at a lower temperature. Additionally, the MCM temperature after demagnetization is higher than that after magnetization, which is

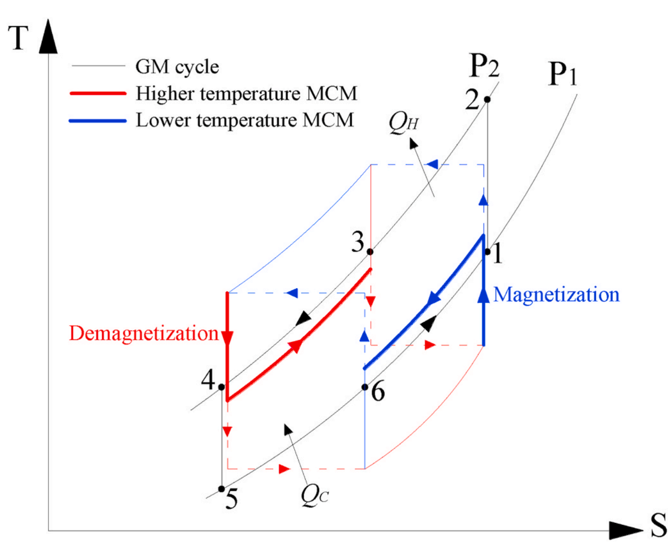



**Fig. 25.** T–S diagram of the hybrid cycle combining the MR cycle and GM cycle (process 1-2-3: compression process; 3–4: hot to cold blow process; 4-5-6: expansion process; 6-1: cold to hot blow process).

attributed to the leading role of the GM cycle in the hybrid cycle.

For the hybrid cycle combining the MR cycle and GM cycle, efforts are made to determine the optimal phase angle. The phase angle is defined as the peak difference between the hot space volume and the magnetic field. Tuning the phase angle enables us to determine the best timing between the MR cycle and GM cycle. In the study by Shen et al. [84], a phase angle of 60° was observed to be best for the hybrid cycle at 0.5 Hz. Here, a minimal no-load temperature of 3.5 K was attained, which was lower than the 4.2 K of the pure GM cycle.

The concept of combining the MR cycle and Stirling cycle is similar to that of combining the MR cycle and GM cycle; therefore, no detailed description is required. For the hybrid cycle combining the MR cycle and Stirling cycle, current research focuses on investigating the effect of the phase angle on the cooling performance. The phase angle is defined as the peak difference between the compression chamber volume and the magnetic field. In the study of He et al. [87], the maximum temperature span occurred at a 90° phase angle when the charged helium pressure was 1 MPa and the operating frequency was 1.5 Hz. Here, the temperature span of the hybrid cycle was 24% higher than that of the pure Stirling cycle at 6 W cooling capacity. Later, Gao et al. [88] determined an optimal phase angle of 60° under the condition of 4.5 MPa charged helium pressure and 1.5 Hz operating frequency. With a 60° phase angle, a lower no-load temperature and a higher cooling capacity were achieved by the hybrid cycle when compared with the pure Stirling cycle.

Based on the above analysis, the phase angle has an important function in the performance of the hybrid cycle combining the MR cycle and gas regenerative refrigeration cycle. An optimal phase angle is conducive to obtaining a positive coupling performance. In contrast, an unreasonable phase angle is most likely to result in a worse performance than that of the pure gas regenerative refrigeration cycle. Therefore, determining the optimal phase angle of the hybrid cycle is a priority.

### 2.4.2. *Multi-caloric refrigeration cycle*

A material with susceptibility to multiple external stimuli (e.g. magnetic field, electric field, stress field, etc.) is a prerequisite to exploiting multi-caloric refrigeration. Multi-caloric refrigeration can be used to achieve a more remarkable total cooling effect, extend the refrigeration temperature region, and overcome the irreversibility of the caloric effect.

To achieve a more remarkable total cooling effect, Czernuszewicz et al. [89] applied a magnetic field and triaxial stress field to a Ni–Mn–In alloy (Fig. 26), combining its MCE and barocaloric effect. To evaluate the multi-caloric refrigeration capacity, the temperature changes of Ni–Mn–In stimulated by a magnetic field and triaxial stress field were directly measured. The results indicated that the temperature change produced by multi-caloric effects was approximately the sum of that produced by the respective effects. Besides, Czernuszewicz et al. [89] concluded that multi-caloric refrigeration could result in a particular temperature change with a 75% lower magnetic field compared with only MR.

To extend the refrigeration temperature region, Hu et al. [90] combined the MCE and elastocaloric effect of a Ni-Mn-Ga alloy ($Ni_{49.5}Mn_{28}Ga_{22.5}$). Specifically, when the Ni-Mn-Ga alloy was stimulated using a magnetic field, there was no MCE in region II from 295 to 330 K (Fig. 27). But coincidentally, when applying the stress, the Ni-Mn-Ga alloy exhibited an elastocaloric effect from 295 to 340 K, which lay exactly in region II. Hence, a broad refrigeration temperature region could be achieved by combining the MCE and the elastocaloric effect.

To overcome the irreversibility of the caloric effect, Liu et al. [91] proposed a magnetic-electric multi-caloric cycle in a multiferroic heterostructure (ferromagnetic FeRh films coupling to a ferroelectric $BaTiO_3$ substrate), as shown in Fig. 28a. Specifically, FeRh could exhibit a giant negative MCE via magnetic field stimuli. However, the MCE in the FeRh was irreversible due to the large hysteresis of its first-order metamagnetic phase transition. Visually, the MCE in FeRh followed a

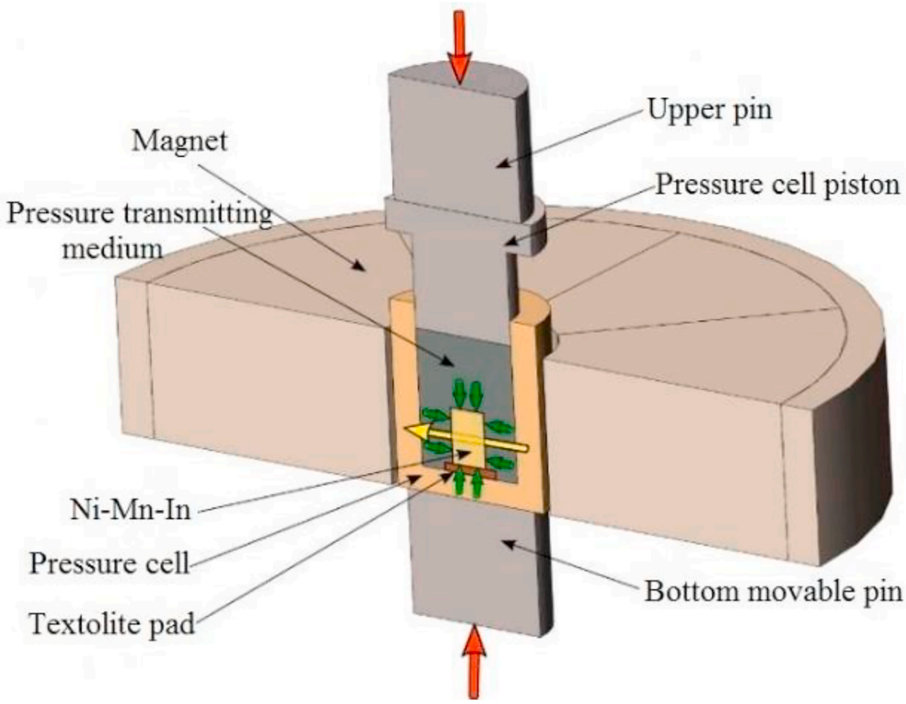



**Fig. 26.** Model of the test system with a magnetic field and triaxial stress stimuli (red arrows: external force; yellow arrow: magnetic field; green arrows: triaxial stress stimuli) [89].

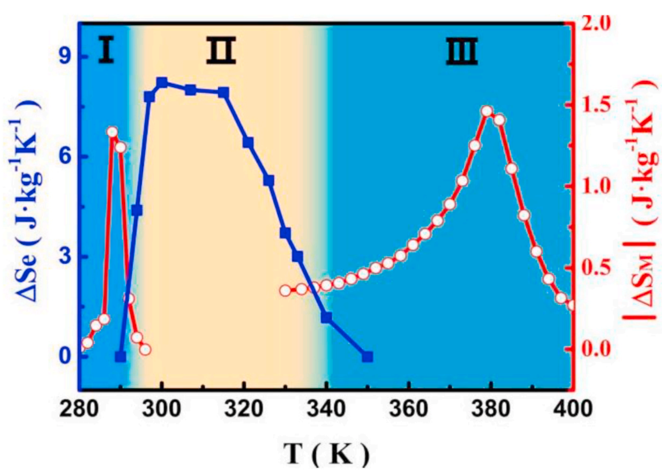



**Fig. 27.** Temperature dependence of multi-caloric effects for Ni–Mn-Ga alloy (region I and III: MCE; region II: elastocaloric effect) [90].

path with a large virgin effect, followed by a smaller reversible pathway (path 1-2-3 in Fig. 28b). Aiming at the hysteresis of transition, Liu et al. [91] indicated that applying an electric field to the $BaTiO_3$ substrate could promote the FeRh transition from the ferromagnetic phase to the antiferromagnetic phase. Accordingly, by selecting a suitable value for the electric field, the hysteresis of the FeRh metamagnetic phase transition could be cancelled, and a reversible MCE was achieved (path 1–4 in Fig. 28b).

As described above, multi-caloric refrigeration is beneficial for obtaining a better cooling performance, including a larger temperature span and broader refrigeration temperature region. However, multi-caloric refrigeration is highly dependent on the material properties, and the discovery of materials exhibiting multi-caloric effects simultaneously is the premise of applying multi-caloric refrigeration.

In summary, combining with other cooling effects provides a new development direction for MR technology. Nonetheless, a keen insight is necessary to determine the entry point at which multiple cooling effects can be combined.

So far various MR thermodynamic cycles have been introduced, including hybrid MR cycle, optimized AMR cycle, fully solid-state MR

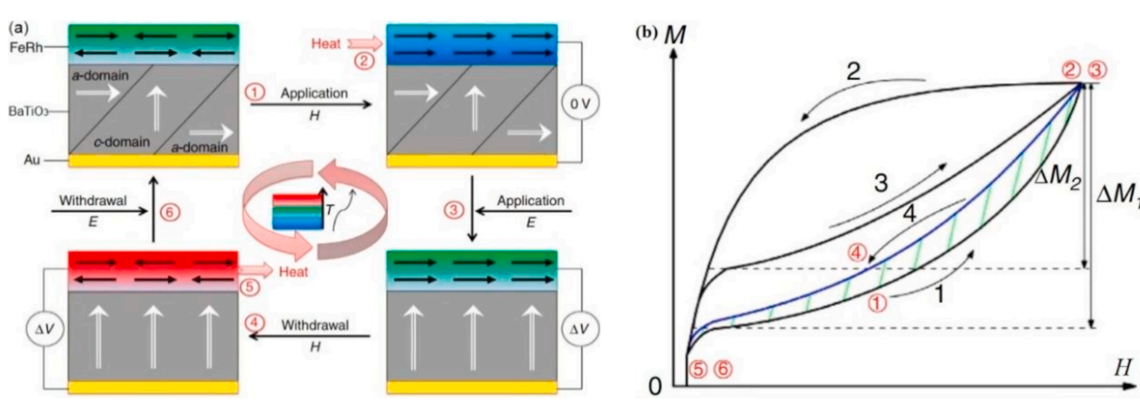



**Fig. 28.** Multi-caloric refrigeration cycle (a) schematic of magnetic-electric multi-caloric refrigeration cycle; (b) magnetization M versus magnetic field H [91].

cycle and MR cycle combined with other cooling effects. Among them, the development of hybrid MR cycle and optimized AMR cycle is relatively mature. The fully solid-state MR cycle and multi-caloric refrigeration cycle are likely to be promising solutions in future. The fully solid-state MR cycle is designed with the ability to rapidly transfer heat from or to the MCM; thus, a high operating frequency is expected to be achieved. Multi-caloric refrigeration cycle combines multiple caloric effects. More caloric effects may result in a greater cooling capacity.

## 3. Heat transfer enhancement during heat regeneration

### 3.1. Methods for enhancing convection heat transfer

Over the past few years, numerous studies have been conducted to enhance the convection heat transfer between the HTF and MCM, regardless of the HTF or MCM.

Regarding HTF, water or water solutions are selected by most magnetic refrigerators. Nonetheless, Kitanovski et al. [47] suggested utilizing liquid metals as HTFs because of their better heat transfer ability. Inspired by this concept, Wu et al. [92] established a 2D numerical model to calculate the cooling capacity of different HTFs. Their results indicated that when using mercury to replace water as the HTF, the cooling capacity could be increased by approximately 600%. However, note that the mercury is highly toxic. Therefore, the practical application of mercury is difficult.

Galinstan, an alloy of 68.5% Ga, 21.5% In and 10% Sn, is considered as a better substitute for mercury in many liquid metal applications owing to its nontoxicity [93]. Lei et al. [94] compared the heat transfer performance of liquid metal Galinstan and water. The results revealed that a no-load temperature span of 27 K could be obtained using Galinstan as the HTF, whereas only 7 K was achieved with water. Kitanovski et al. [95] investigated the performance of a magnetic refrigerator using two different HTFs (liquid metal Galinstan and 20% ethanol-water). Their results indicated that a better cooling performance, including cooling capacity and COP, could be obtained with Galinstan. In conclusion, liquid metal Galinstan can be regarded as a promising HTF for applications in MR.

Table 4 shows some common room-temperature liquid metals (including mercury and Galinstan: 68.5% Ga, 21.5% In, 10% Sn) and their melting temperatures. As shown in Table 4, most of these liquid metals are Ga-based alloys. However, one disadvantage of Ga-based alloys is that as soon as the Ga-based alloys are exposed to ambient air, their surface will be oxidized immediately (far less than 1s) [96]. This disadvantage makes Ga-based alloys easily oxidized to high viscosity oxides ($Ga_2O_3$, $Ga_2O$, etc. [96]) if the MR system is not completely sealed. Furthermore, these high viscosity oxides will result in an increase in the pressure drop and pipe blockage. Therefore, if the application of Ga-based alloys is considered, it is necessary to try to prevent them from being oxidized.

Nowadays, nanomaterials are widely utilized in various fields, such as the environment field [101,102] (especially for the wastewater treatment [103–105]) and biomedical field [106,107]. In addition, it has long been recognized that suspensions of nanomaterials in liquids (known as nanofluids) have great potential in enhancing heat transfer [108]. As reported by Saidur et al. [109], adding 0.4% nanoparticles to

**Table 4**
Room-temperature liquid metals and their melting temperature.

| References | Liquid metals | Melting temperature/°C |
|---|---|---|
| [97] | mercury | −38.8 |
| [98] | Galinstan: 68.5% Ga, 21.5% In, 10% Sn | −19 |
| [99] | Galinstan: 62.5% Ga, 21.5% In, 16% Sn | 10.7 |
| [99] | EGaIn: 75.5% Ga, 24.5% In | 15.7 |
| [99] | Ga–In: 78.6% Ga, 21.4% In | 15 |
| [100] | Ga-In-Sn-Zn | 9.7 |
| [99] | K–Na: 76.7% K, 23.3% Na | −12.7 |

the initial liquid can increase its thermal conductivity by 40%. Thus, nanofluids as HTFs should be considered promising [110]. Nanofluids are categorized into conventional nanofluids (e.g. copper-water nanofluids) and magnetic nanofluids. Magnetic nanofluids consist of a suspension of ferromagnetic nanoparticles (e.g. Fe, Co, and Ni) into nonmagnetic carrier fluids. These ferromagnetic nanoparticles enhance the thermal conductivity of nanofluids and offer more phenomenal features owing to their magnetic nature [111]. Considering using magnetic nanofluids as HTFs, the flow of the HTF can be induced by the movement of the magnet, which can avoid the use of a reciprocating pump. Furthermore, if nanoparticles are MCMs, nanofluids will be magnetocaloric nanofluids exhibiting MCE. Fig. 29 shows a schematic diagram of the MR device with magnetocaloric nanofluids [112]. In such an MR device, particle concentration is crucial. To obtain a high cooling capacity, a high particle concentration is expected. However, to guarantee a homogeneous distribution of particles and to prevent particle clustering, the particle concentration must be limited to a certain value. Hence, developing an MR device with magnetocaloric nanofluids is difficult. To date, there have been no reports on MR prototypes using magnetocaloric nanofluids.

With respect to MCMs, Gd is the most commonly employed one for room-temperature MR [114]. This is attributed to its good MCE, large operating temperature zone, low hysteresis, and good mechanical properties. Other commonly used ones are LaFeSi-based and MnFeP-based alloys [113]. Note that every MCM has a Curie temperature at which the MCE is maximum. As the MCM temperature deviates from its Curie temperature, its MCE decreases. A solution to this problem is to use multi-layered MCMs with different Curie temperatures along the length of the AMR [115]. The MCM with the lowest Curie temperature is placed at the cold end, and the MCM with the highest Curie temperature is placed at the hot end (recent studies are listed in Table 5 [116–120]). Such a design enables each MCM layer to experience maximum possible temperature increase and decrease, thereby enhancing heat transfer performance from the perspective of increasing the heat transfer temperature difference. However, for a particular AMR length, the number of MCM layers must be optimized. For a particular number of MCM layers, the appropriate Curie temperature difference should be explored.

It is clear that the heat transfer performance depends not only on the MCM nature but also on the MCM geometry. Currently, the geometries of MCM are focused on spheres and plates. However, the AMR filled with sphere-MCMs has a large pressure drop owing to its irregular flow path, and the AMR filled with plate-MCMs has a much smaller specific heat transfer area [121]. To address these limitations, an effective solution is to shape the MCM into a micro-channel structure with a lower pressure drop and a larger specific heat transfer area, i.e. micro-channel AMR. Unlike the micro-channel heat exchanger, the channel shape of the micro-channel AMR can be rectangle or circular, but the diameter of circular channels should not be less than 0.1 mm [122]. For the micro-channel AMR (Fig.30a), a performance comparison with

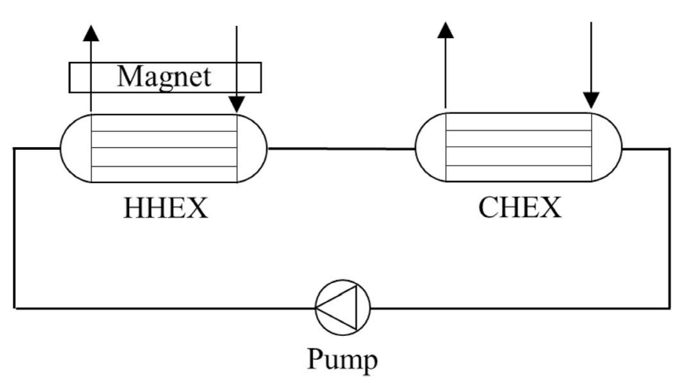



**Fig. 29.** Schematic diagram of an MR device with magnetocaloric nanofluids.

**Table 5**
Research progress of multi-layered MCMs in recent five years.

| References | Layer number | MCM | Measures for adjusting the Curie temperature | Range of Curie temperature |
|---|---|---|---|---|
| [116] | 4 | $La(Fe,Co)_{13\text{-}x}Si_x$ | Changing Si fraction | – |
| [117] | 2 | Gd, $Gd_{0.73}Tb_{0.27}$ | Changing Tb fraction | 277–294 K |
| [118] | 16 | La–Fe–Mn–Si–H | – | 266–307 K |
| [119] | 9 | $La(Fe,Si,Mn)_{13}H_y$ | Changing H fraction | 284.5–299.2 K |
| [120] | 2 | $Mn_{5-x}Ge_3Ni_x$ | Changing Ni fraction | 288–293 K |

parallel-plate AMR (Fig. 30b) was performed by Kamran et al. [123] for the same MCM mass and overall dimensions. The results indicated that the maximum cooling capacity of the micro-channel AMR was 7% higher than that of the parallel-plate AMR, and the cooling capacity per unit pumping power was the highest when using a micro-channel AMR. Furthermore, Kamran et al. [124] employed multi-layered MCMs along the length of the micro-channel AMR to further improve the AMR performance. In addition, in terms of the manufacture of micro-channel AMRs, although some shaping methods, such as selective laser melting [125] and additive manufacturing method [126] exist, shaping the MCM into a micro-channel structure is difficult owing to the limitations of the mechanical properties of MCMs.

The mentioned micro-channel AMR can enhance the heat transfer performance by increasing heat transfer area. Moreover, some interest has been devoted to increasing the heat transfer coefficient. Chen et al. [127] presented an enhanced heat transfer structure of a staggered mini-plates type for application in MR (Fig. 31b) and compared its heat transfer characteristic with that of flat plate type (Fig. 31a). The results indicated that a relatively high heat transfer efficiency could be realized with the flat plate type replaced by staggered mini-plates type, and the highest increment of heat transfer efficiency was ~8%. Moreover, considering the pressure loss, the heat transfer efficiency of both types decreased, but the staggered mini-plates type had a larger decrease. Thus, the effect of pressure loss on the heat transfer performance must be considered, particularly for enhanced heat transfer structures. You et al.

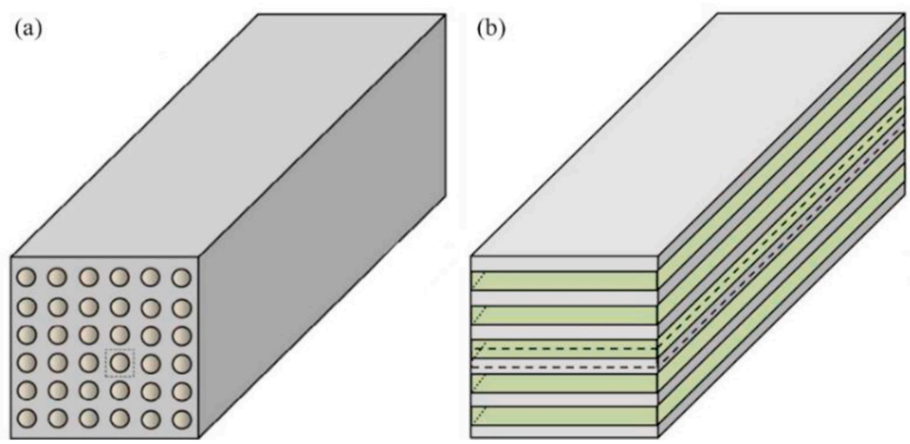



**Fig. 30.** Geometries of AMR (a) micro-channel AMR; (b) parallel-plate AMR [123].

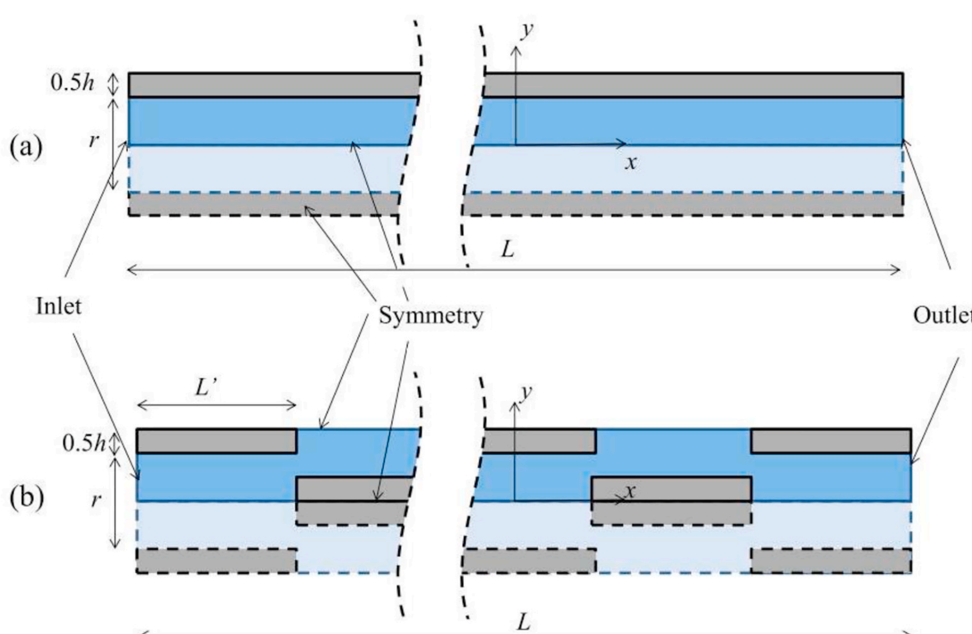



**Fig. 31.** Two types of flow channels (a) flat plate type; (b) staggered mini-plates type [127].

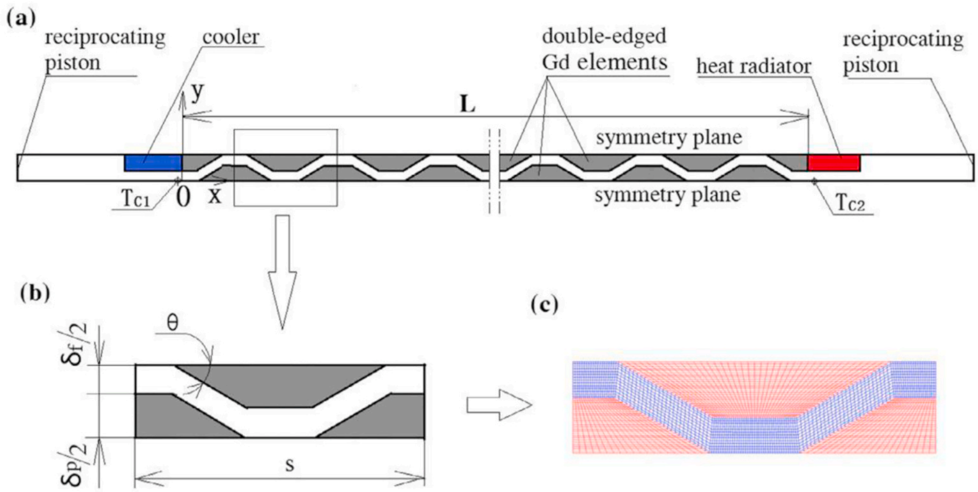



**Fig. 32.** Unit structure of a magnetic refrigerator with twin-wedged Gd elements (a) unit structure of magnetic refrigerator; (b) local zoom for plate arrangement; (c) local meshes generation [128].

[128] compared the performance comparison of magnetic refrigerators with twin-wedged Gd elements (Fig. 32) and long Gd plates. The results revealed that by using twin-wedged Gd elements, a notable heat transfer enhancement and an increase in the temperature span, SCP, and COP could be achieved in comparison with using long Gd plates. Nonetheless, when adopting the twin-wedged Gd elements in a magnetic refrigerator, the flow resistance and pressure loss would increase since the flow boundary layer was redeveloped periodically. To summarize, the enhanced heat transfer structure has a strong disturbance to the HTF, thereby improving the heat transfer coefficient between the MCM and HTF. However, the large pressure loss caused by complex structures cannot be ignored in practical applications.

In summary, regarding the HTF, liquid metals and nanofluids are considered to be promising alternatives for water or water solutions. Regarding the MCM, using multi-layered MCMs with different Curie temperatures along the length of AMR can enhance convection heat transfer from the aspect of increasing temperature difference, but two key parameters (the number of MCM layers and the Curie temperature difference) are required to be optimized. The micro-channel AMR can enhance convection heat transfer by increasing the heat transfer area, but MCMs with good mechanical properties and simple shaping methods are worth exploring. The enhanced heat transfer structure can enhance the convection heat transfer from the aspect of increasing the heat transfer coefficient, but a large pressure loss must be considered.

### 3.2. Methods for enhancing heat conduction

In the fully solid-state MR cycle, the heat conduction enhancement is well reflected in the study of Lu et al. [75]. On one hand, they inserted a certain volume fraction HTCM in the MCM to compensate for the limitations of the low thermal conductivity of MCM. To confirm the optimal distribution of a HTCM (e.g. copper) in the MCM (e.g. Gd), topology optimization was performed with a limit of a 0.66 Gd volume fraction. The optimization process and final topology optimization structure is shown in Fig. 33. Regarding this topology optimization structure, Lu et al. [75] compared its heat transfer behavior with that of the other two heat transfer structures (parallel sheets and Gd-only, Fig. 18). The results indicated that the best heat transfer capability was obtained by the topology optimization structure, followed by parallel sheets and Gd-only. It worth noting that the optimized heat transfer structure would vary with the defined conditions, such as the size, shape, and MCM volume fraction. However, the topology optimization method and the concept of enhancing heat conduction can serve as guidelines for future studies. In addition, the shaping difficulty of the optimized heat transfer structure and air thermal resistance between the MCM and HTCM should be considered in practical applications.

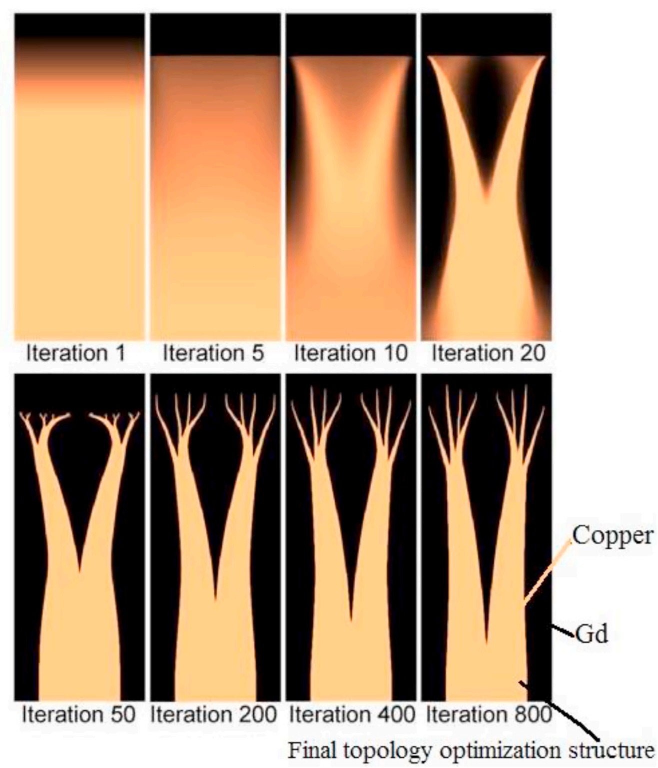



**Fig. 33.** Topology optimization process [75].

On the other hand, Lu et al. [75] utilized Peltier elements instead of copper blocks as middle heat transfer blocks to enhance heat regeneration between paired lattices. Peltier elements contribute to enhancing heat conduction in all fully solid-state MR cycles based on Peltier elements (described in Section 2.3.1), not only in the MUR cycle proposed by Lu et al. [75]. However, two problems, Joule heating and heat leakage, must be overcome when applying Peltier elements. In addition, with the development of heat pipe technology and materials science, it is possible to utilize the vapor chamber or graphene as the heat transfer medium between paired MCMs.

Based on the above analysis, we can conclude that enhancing heat transfer during heat regeneration is essential for improving the magnetic refrigerator performance. The heat transfer enhancement can shorten the operating period of each MR cycle, and thus increase the operating frequency of magnetic refrigerators. However, several problems must be solved, such as the poor mechanical properties of MCMs, difficult shaping of complex structures and, large pressure loss caused by enhanced heat transfer structures.

## 4. Application research of room-temperature MR

### 4.1. Application in cold-storage device

Over recent years, several institutes have participated in the research and development of cold-storage devices based on MR technology, including the General Electric (GE) Company, Haier, Baotou Research Institute of Rare Earths (BRIRE), and Cooltech Applications.

In 2014, GE declared that they had developed a silent MR refrigerator that could cool a bottle of beer, and this novel refrigerator was expected to appear on the market within a decade [129].

In 2014, Haier announced that they had successfully developed an MR technology that could be applied to the civilian field, and this technology was first applied to wine coolers. In 2015, Haier exhibited a wine cooler without a compressor at the Consumer Electronics Show (CES) [130], as displayed in Fig. 34.

In 2015, BRIRE designed an MR cold-storage cabinet with a refrigeration space of 117 L and a maximum refrigeration temperature difference of 26 °C, which could cool canned beers from 25 to 5 °C. Moreover, BRIER developed two commercial MR cold-storage cabinets (Fig. 35), whose magnetic field was provided by a permanent magnet (1.3 T), and the MCM was Gd and its alloy with a diameter of 0.4–0.6 mm. In addition, these prototypes with 110 L refrigeration space could achieve a maximum refrigeration temperature difference of 22.7 °C, and maintained the cold-storage room temperature at 5–10 °C.

In 2015, Cooltech Applications exhibited a medical refrigerator based on MR at MEDICA [131]. In 2016, they introduced a commercial MR refrigerator (type MRS400), which could maintain the temperature inside the refrigerator at 2–5 °C using a 400 W cooling power [132]. Such a 2–5 °C temperature range could satisfy the requirements of food preservation.

In 2021, the Federal University of Santa Catarina [67] developed a magnetic wine cooler prototype, whose cabinet could hold up to 31 bottles of wine. This wine cooler could maintain a mean temperature of 10.8 °C inside the cabinet for a 25 °C ambient temperature.

Based on the above description, we can conclude that the development of cold-storage devices based on MR has progressed significantly; therefore, we consider that various MR cold-storage devices, such as MR wine coolers and refrigerators, will be available in the market in the near future.

**Fig. 34.** Wine cooler exhibited by Haier at CES.

**Fig. 35.** Two MR cold-storage cabinets developed by BRIRE.

In addition to the cold-storage device, other small cooling devices can also be the primary application targets that MR technology aims at, including the domestic dehumidifier and portable personal air conditioning [133].

### 4.2. Application in heat pump

At present, a single magnetic refrigerator or magnetic heat pump cannot be implemented in buildings owing to its insufficient temperature span and cooling or heating capacity. However, Johra et al. [134, 135] developed a ground source magnetic heat pump system to provide the heating requirements of a building, which provided a new concept for the application of magnetic heat pump. Fig. 36 shows the schematic diagram of the ground source magnetic heat pump system. The magnetic heat pump was integrated in a single hydronic loop with a ground source heat exchanger (GSHE) and a floor heating system. Compared with the conventional ground source heat pump system (Fig. 37), there was no intermediate heat exchanger or hot water storage tank. This configuration has the potential to decrease the temperature difference between heat source and heat sink, simplifies the piping network, and reduces heat losses. For this novel ground source magnetic heat pump system, Johra et al. [134] demonstrated the possibility of applying it in a low energy single family house for the first time through numerical simulation and concluded that the magnetic heat pump could deliver 2600 W of heating power with a 3.93 average seasonal system COP. Here, the magnetic heat pump was only utilized to heat the living room. However, when the magnetic heat pump was utilized to heat the entire house with multiple thermal zones regulated by simple on-off controllers, a poor average seasonal COP of 1.84 was obtained.

Aiming at the poor heating performance, Johra et al. [135] indicated that the heating requirement profiles in each room varied owing to different internal people load, equipment load, and solar load. Thus, it was almost impossible for all rooms to require maximum heating power simultaneously. As a result, the single control strategy (the valve was fully open when the room temperature was below the set-point value and the valve was fully closed when the room temperature was above the set-point value), as adopted by the former study [134], would lead to a low system COP. To improve the overall performance of the heat pump system, a new control strategy using the building energy flexibility potential was implemented. The results indicated that a higher COP (2.91, 3.48, and 3.51 for light, medium, and heavy structure houses, respectively) could be achieved. However, the stability of room temperature

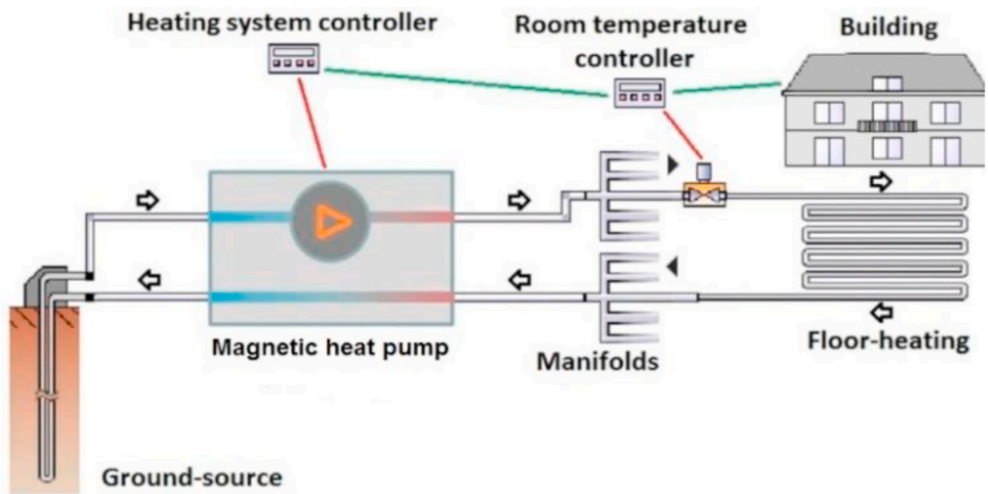



**Fig. 36.** Ground source magnetic heat pump system [134].

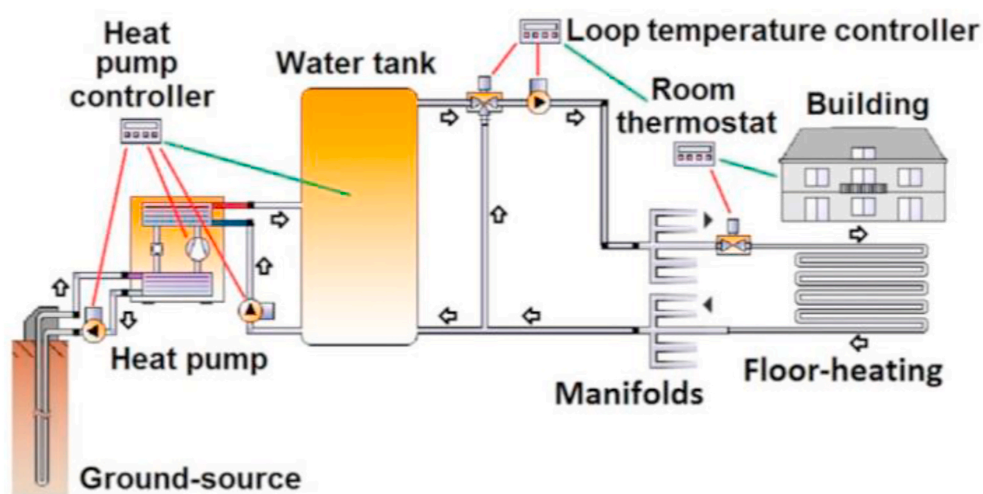



**Fig. 37.** Conventional vapor-compression ground source heat pump system [134].

was reduced.

Afterwards, if the feasibility of coupling a magnetic heat pump and a GSHE together can be experimentally verified, the application field of MR technology will be broadened. Moreover, in addition to geothermal energy, Han et al. [136] coupled the solar energy into MR technology and invented a solar MR seawater desalting device. In this device, the solar energy was utilized to heat seawater and generate steam. The MR was utilized to condense the steam and catch the fresh water. In the future, more other renewable energy (e.g. biomass energy) can also be considered for coupling to MR technology.

### 4.3. Application in electric vehicle air conditioning

In electric vehicles, the air conditioning system provides a comfortable thermal environment for passengers. However, because there is no waste heat available from the electric vehicle and the lithium-based battery also needs to be cooled, the requirements for heating and cooling for electric vehicles are greater than those of thermal motor vehicles. Furthermore, these required heating and cooling capacity are produced with battery in electric vehicles, which leads to a significant decrease in mileage. Hence, to reduce the power consumption of heating and cooling, it is a good choice to apply MR technology in electric vehicles.

Noume et al. [137] established an AMR cycle model intended for designing an MR system for an electric vehicle and determined the velocity field, temperature field, convective heat transfer coefficient, and system temperature span. However, this study focused on the behavior of the AMR cycle, and did not consider the special application scenario of electric vehicles. As an improvement, Torregrosa-Jaime et al. [138] developed an overall model of the magnetic air conditioning system of an electric minibus, including thermal models of the cabin, magnetic heat pumps, and thermal power distribution loops. Based on the analysis of this overall model, Torregrosa-Jaime et al. [138] reported that a cooling power of 1.6 kW over a temperature span of 37 K was required in the cooling mode, and a heating power of 3.39 kW over a temperature span of 40 K was required in the heating mode. Besides, the magnetic air conditioning system consisted of two magnetic heat pumps, one in the driver's region and the other in the passengers' region (Fig. 38). Such a layout was beneficial for achieving different thermal comfort conditions in different regions. Also, attaining the required cooling/heating power of the electric minibus seemed much easier if two magnetic heat pumps were installed instead one.

In summary, the conclusion obtained by Torregrosa-Jaime et al. [138] can serve as a reference for the magnetic air conditioning system design of an electric vehicle. But to the best of our knowledge, current room-temperature MR prototypes can produce cooling capacity up to 3024 W at zero-temperature span [6], or temperature span as high as

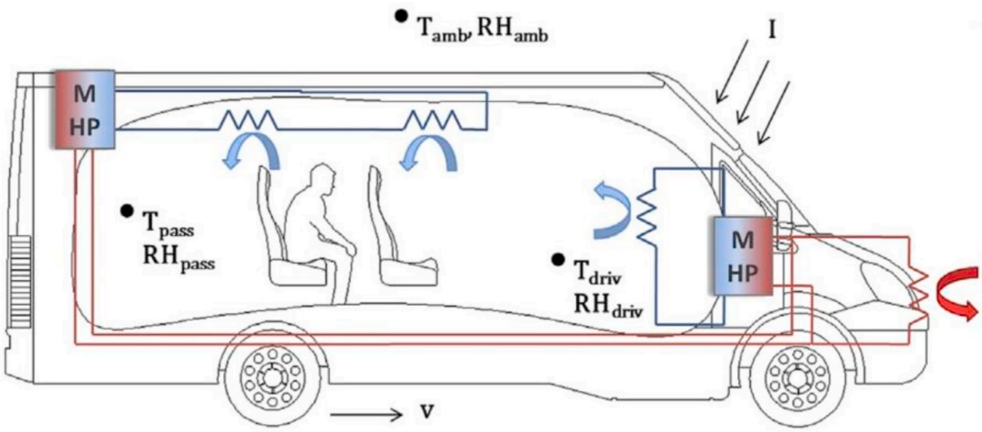



**Fig. 38.** Layout of the magnetic air conditioning system of the electric minibus [138].

around 40 K under zero-load condition [20], which are still below the requirements for the application of electric vehicles. Therefore, much development is still required before MR technology is applied to electric vehicles.

Overall, limited by the temperature span and cooling capacity, room-temperature MR is first considered to be applied to some cooling devices that require less cooling capacity.

## 5. Conclusions

Room-temperature MR is considered as a promising alternative to VC refrigeration due to its environmentally friendly and efficient features. Nowadays, the temperature span and cooling capacity obtained by MR prototypes cannot satisfy the requirements for practical cooling. To improve the temperature span and cooling capacity, solutions are primarily developed from two aspects: MR thermodynamic cycles, and heat transfer enhancement during heat regeneration.

In terms of MR thermodynamic cycles, hybrid MR cycle, optimized AMR cycle, fully solid-state MR cycle, and MR cycle combined with other cooling effects are established. The hybrid MR cycle can combine the high temperature span of the magnetic Brayton cycle and the high cooling capacity of the magnetic Ericsson cycle, thereby achieving a higher cooling capacity at a higher temperature span. The optimized AMR cycle reduces the irreversible loss by the optimal design of fluid circuit. The fully solid-state MR cycle is characterized by high operating frequency (over 200 Hz) and high SCP, but only numerical simulation study is conducted. The MR cycle combined with other cooling effects can obtain a better cooling performance by the simultaneous action of multiple cooling effects. Among them, the development of hybrid MR cycle and optimized AMR cycle is relatively mature. The other two cycles are likely to be the main direction of the future research.

With respect to heat transfer enhancement during heat regeneration, three methods, including using liquid metals or nanofluids as HTFs, shaping MCMs into enhanced heat transfer structures, and inserting HTCM in MCM can theoretically shorten the heat transfer time (operating period), and improve the operating frequency. However, these three methods have not yet been applied in MR prototypes.

In the future, research on room-temperature MR technology is best to focus on the experimental study. Specifically, a fully solid-state MR prototype should be built to experimentally verify the feasibility of fully solid-state MR cycle. Besides, when three enhanced heat transfer methods are put into practice, several problems must be solved, such as the oxidation of Ga-based liquid metals, the aggregation of nanoparticles, and the difficulty in processing enhanced heat transfer structures.

In addition, room-temperature MR application in cold-storage device, heat pump system and electric vehicle air conditioning are reviewed in this paper. Limited by the development of MR technology, room temperature MR is first considered to be applied to devices with small cooling capacity, such as the wine cooler, domestic dehumidifier, and portable personal air conditioning.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgments

This work was supported by the National Natural Science Foundation of China [Grant No.51776076] and the Natural Science Foundation of Guangdong Province (cultivation of major basic research) [Grant No.2018B0308006].

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